

Chapter 7

Plasma Stripping, Cleaning, and Surface Conditioning

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**7.1 INTRODUCTION TO PLASMA STRIPPING AND
CLEANING**

Plasma processing for integrated circuit (IC) manufacturing is ubiquitous. Typically, a wafer is processed through multiple plasma processing steps before it emerges from the fab as an array of ICs. Plasma processes include thin film deposition, etching, photoresist stripping, and cleaning, also known as residue removal. Cleaning processes prepare the wafer surface chemically and in some cases structurally for the next step. This chapter focuses on photoresist stripping, cleaning processes, and applications. The chapter will also touch upon other applications such as soft-silicon etching, atomic layer etching (ALE), pre-deposition surface cleaning including cleaning for epitaxial film growth, and chamber cleaning to remove films and residues.

**7.1.1 Overview of Plasma Stripping, Cleaning, and Surface
Conditioning Applications**

In state-of-the-art IC fabrication processes, more than 150 cleaning steps are invoked; see Chapter 1 for an overview of all cleaning process steps. Plasma

stripping and cleaning are used for a variety of applications and serve a variety of purposes. The most common of these applications are as follows:

Stripping: remove bulk photoresist materials (also referred to as resist) after either plasma etch or ion implantation steps

Residue Removal: remove residues at the bottom or sidewalls of an etched via or trench that result from the ablation of underlying barrier, etch stop, or metal layers; remove polymer or inorganic residues that are formed to establish selectivity to underlying layers and to generate anisotropic profiles

Pre-deposition Cleaning: clean surfaces of films, residues, and particles before dielectric or metal film deposition to ensure good adhesion or low contact resistance

Surface Conditioning: alter or condition the surface to inhibit oxidation and change surface film composition or adsorption of impurities during process sequences; in addition, change the surface tension to facilitate subsequent wet or dry process.

In most cases, the stripping or cleaning cycle must meet two or more of these process requirements. [Fig. 7.1-1](#) shows examples of the difference between stripping bulk photoresist material and residue removal.

Surface conditioning is essential before film deposition or wet processes, in particular epitaxial deposition of Si, SiGe, or other materials that are to be deposited on the crystalline Si surface. Chapter 9 reviews the requirements for and the production of an H-terminated Si surface.

Plasmas are used routinely to clean deposited films or residues from vacuum chambers after plasma etch, plasma deposition, and chemical vapor deposition (CVD) processes. Plasmas are also used to condition the inside of the processing chambers before the deposition process; that is, plasmas can release or etch away potential particles or remove material from the reaction chamber that may affect process controllability and subsequently deposit a protective passivating film. In single-wafer etch sequences, a chamber clean is often performed after the etch step to ensure reproducible chamber conditions for the following wafer etch cycle. Although the required precision and control are not as stringent as those of photoresist stripping and surface preparation, these cleaning applications may use large volumes of gases. As a result, environmental concerns and cost are often paramount and will be discussed in future sections.

7.1.2 Introduction to Plasma Processing

The reactive chemical atmosphere in a glow discharge or plasma offers the potential to effectively modify and/or clean a solid surface at low temperatures and high rates. Impact collisions between high-energy electrons and feed gas molecules create reactive chemical moieties such as free radicals and ions in

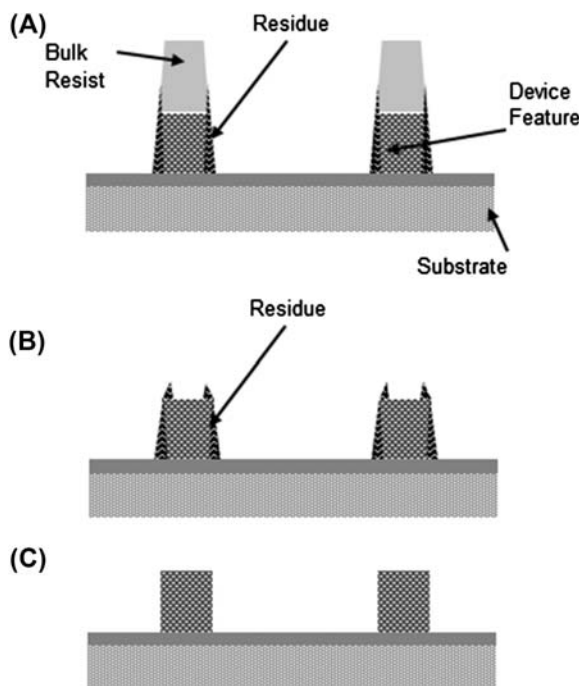


FIGURE 7.1-1 Schematic of device feature showing the difference between stripping and cleaning. Stripping removes the bulk photoresist, but generally leaves residue. Cleaning removes the residue remaining on the feature. The schematics are: (A) after etch and before stripping and cleaning, (B) after stripping and before cleaning, and (C) after stripping and cleaning.

ground states and in excited states. If subsequent reactions between these chemical species and films or residues on wafer surfaces result in volatile products, film/residue removal occurs. Alternatively, surfaces can be modified or conditioned by plasma reactions to alter the chemical or physical nature of the surfaces. However, for plasma cleaning and stripping processes to be viable, acceptable selectivity to existing films or surfaces must be established, and radiation damage to or alteration of existing films must be minimized. In principle, these results can be achieved without the use of large quantities of aqueous acids and bases, organic solvents, or deionized H_2O , thereby establishing relatively environmentally benign processes. However, in many post-stripping applications, a liquid treatment is used in conjunction with a plasma stripping process for complete residue removal, especially when inorganic metal residues are present.

A plasma is a partially ionized gas formed by coupling radio frequency (RF) or microwave (MW) energy into a low-pressure gas. Collisions between electrons that have been accelerated by the applied electric field and neutral gases or vapors result in dissociation of the gases and vapors. The plasma

consists of a variety of chemical species in different electronic states, approximately equal densities of negatively and positively charged species and radicals, stable neutral species, and free electrons. Sufficient energy must be imparted to the plasma to sustain this high-energy gas, because recombination occurs through collisions with the walls of the reaction vessel and with other species in the plasma. Because of the high-energy, highly reactive nature of the plasma, enhanced chemical reactivity is achieved with films and the wafer surface relative to what would be observed under ambient conditions. The plasma parameters (e.g., electron energy distribution, ion energy distribution, residence time) that control the ionization, dissociation, and excitation in plasmas are dependent on the pressure, the gases used to form the plasma, the plasma frequency, and the power (or power pulsing) applied to the plasma. The gas used for cleaning must dissociate to form chemically active species that react with the films to be removed but these reactive species should not react with the underlying films. Selection of appropriate gases and plasma conditions ensure that the cleaning process can etch various materials, strip photoresist, and remove post-etch and implant residue and condition the wafer surface by H-passivating the Si and removing native oxide. For example, an O_2 plasma generates O radicals, electrons, positively and negatively charged oxygen species (e.g., O_2^- , O^- , O_2^+ , O^+), and excited state species. The excited species are molecules or atoms that have absorbed energy and enter a quantum state that is at a higher energy than the ground state of the species. Thus, the various excited and dissociated oxygen species can remove photoresist at a much faster rate than can be achieved in an O_2 ambient environment—where essentially no reaction takes place.

7.1.3 Challenges of Plasma Stripping, Cleaning, and Surface Conditioning

Plasma stripping, cleaning, and surface conditioning are used for many different film layers when fabricating an IC. The layers are composed of various materials, and these different materials require stripping and cleaning processes suitable for the specific materials. The cleaning process must remove all undesirable material and must not deteriorate the underlying films, or yield will suffer. The temperature of the process, the reactant gas mixture, or the ion flux must not negatively affect microelectronic device characteristics. A cleaning process suitable at one stage in wafer processing may not suffice at another because of the specific residues on the surface, which may require tailored cleaning chemistries and processes. These cleaning procedures must be reproducible. In addition, contamination must be controlled during the process because addition of particles or metallic contamination may adversely affect the electrical properties of the device. Furthermore, thermal budgets for the overall manufacturing process must be controlled as device dimensions shrink; this is especially important for shallow junctions.

Difficulties and limitations involved in photoresist stripping and surface cleaning, such as cross-contamination between layers of different materials depend upon whether the process takes place at the front end of line (FEOL) before gate dielectric formation, FEOL source/drain post-implant, middle of line (MOL) during formation of the gate structure and first level interconnect, or back end of line (BEOL) after via, trench, or metal etch. With the advancement of metal gate technology below 16-nm logic technology, for example, the MOL also requires separation of different materials to prevent cross-contamination. Each application has its own unique set of challenges as outlined in the International Technology Roadmap for Semiconductors [1]. For FEOL pregate cleaning, oxides and carbonaceous residues must be removed without roughening or having significant material loss at the Si or SiGe surface. Removal of post-ion implant photoresist layers presents a formidable challenge, particularly for high-dose implants, where tenacious, graphitelike residues must be selectively removed with no damage to underlying layers. For BEOL, sidewall and trench/via bottom residues comprising organic and inorganic components must be removed, again with little or no damage to existing structures or film compositions. For MOL and BEOL metal deposition, the surface on which the metal is deposited must be free of oxides to allow low resistance between the existing metal layer and the deposited metal layer. For MOL, the surface of compound semiconductors must not be eroded or oxidized. Although the reactive plasma atmosphere of ions, electrons, and photons that bombard surfaces is conducive to the rapid removal of unwanted films, residues, and the establishment of anisotropic profiles, inherent disadvantages exist. For instance, chamber and fixture surfaces are typically bombarded by high-energy particles, thereby causing sputtering of impurities onto wafer surfaces. Similarly, impurities in the plasma atmosphere are ionized and can be implanted into existing or growing films or substrates. Large ion and electron currents that flow in high-density reactors can damage dielectrics by breaking chemical bonds during current flow and by dielectric charging ultimately degrading device reliability or leading to dielectric breakdown. Such concerns must be eliminated or at least mitigated by controlling plasma properties or by post-plasma treatments.

One of the most challenging photoresist stripping problems that has emerged in recent years involves the use of low-dielectric constant (low-k) materials in BEOL processing sequences. Because such dielectrics are either polymeric, contain carbon, and/or have porosity, exposure to O- or F-containing plasma atmospheres removes C from the film and causes an increase in k value; ion bombardment can also result in film densification, again increasing k value. Although reducing (H₂- and N₂-containing) plasmas avoid strongly oxidizing chemistries, removal rates and efficiency of stripping can be compromised. Typically, the post-low-k dielectric etch stripping step is performed in situ inside the same plasma chamber as the etching step. However, fluorine memory, the release of etching gases, or fragments adsorbed

on to the chamber walls induce low-k material damage if the stripping and etching processes are performed in the same chamber. Therefore, it is desired to move the wafer to a dedicated stripping chamber free of fluorine-containing residue to reduce such damage.

Considerable effort has been expended to develop plasma processes for this application, mostly involving downstream approaches to minimize radiation damage and sputtering of impurities onto sensitive surfaces. Issues such as process optimization, abatement, and recovery/recycle of perfluorinated compounds (PFCs) for plasma-based cleaning of chambers are beyond the scope of this review. Selected discussion of these issues can be found in other references [2,3], where the phase-out of PFCs for etching, stripping, and cleaning processes is also discussed.

7.2 APPLICATIONS OF PLASMA STRIPPING, CLEANING, AND SURFACE CONDITIONING

The applications for plasma stripping and cleaning are numerous. During the IC manufacturing process, stripping and cleaning of the surface takes place after every etch or implant step, cleaning takes place before deposition, after chemical mechanical polishing (CMP) steps, in some cases after the deposition step, and also in pre-lithography steps where surface contamination is a concern.

7.2.1 Overview of Plasma Stripping, Cleaning, and Surface Conditioning Applications

Fig. 7.2-2 shows a typical cleaning process used in IC manufacturing that incorporates plasma stripping steps. After ion implant or etching steps that utilize photoresist for masking, stripping of the photoresist and cleaning of any

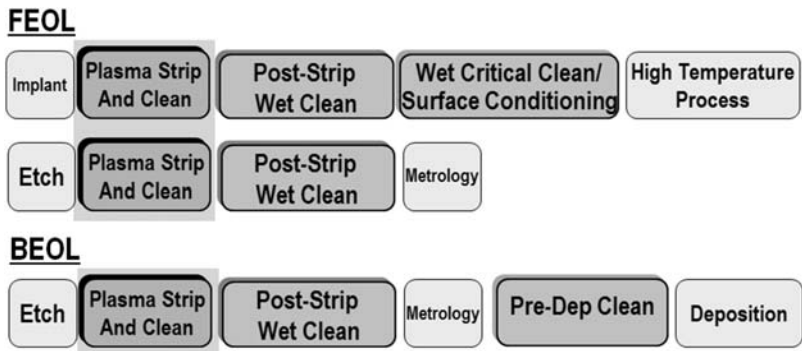


FIGURE 7.2-2 Typical process flows utilizing plasma strip and cleaning steps. Note, most plasma strip and clean steps are followed by a wet clean step.

FEOL and BEOL

FIGURE 7.2-3 Typical process flow utilizing plasma predeposition clean steps. The plasma predeposition clean is typically preceded by a wet cleaning step. In most cases, the plasma predeposition clean step is in situ before the deposition step.

post-strip residue are required. Plasma stripping is the method of choice for removing the photoresist because of a variety of reasons that will be discussed in subsequent sections of this chapter. The post-strip cleaning is usually an aqueous, semiaqueous, or solvent-based process. See Chapter 4 for an overview of wet cleaning processes.

Integration of plasma photoresist stripping with wet cleaning and polymer removal after etching processes is discussed by Martinetti et al. [4]. Typically, the post-etch cleaning process involves the use of plasma photoresist strip steps followed by a wet cleaning step. However, plasmas are also employed as a post-strip cleaning step, to remove unwanted polymer, and in some cases to remove damaged Si or undesirable sputtered metals or SiO_x materials.

Fig. 7.2-3 shows a typical remote plasma processing sequence for Si wafer cleaning. Pre-deposition cleaning of high-density plasma oxide (HDP), epitaxial Si or Si/Ge, and pre-Al or Cu barrier deposition has involved H_2 -based remote plasma cleaning. As previously noted in Chapter 4, pre-diffusion and oxidation processes use wet cleaning chemistries to minimize contaminants on the Si surface before introduction into high temperature or vacuum environments. The wet chemistry has usually involved a modified RCA treatment of SC-1 or SC-2 (SC is standard cleans) followed by a dilute HF immersion and deionized water (DIW) rinse and surface tension gradient dry.

7.2.2 Bulk Photoresist Stripping

The primary objective in bulk resist stripping is the rapid removal of resist material without etching or damaging the underlying films or substrates. Oxygen-based plasma removal of resist (organic) materials appears to proceed in a manner analogous to the thermal oxidation of polymers [5,6]. However, the reactive free radicals and excited state species present in plasmas enhance chemical reactions with the polymer, thereby increasing the removal rate. For instance, O atoms abstract H from the polymer, forming H_2O and OH, while creating highly reactive C radicals in the polymer material for reaction with O species to readily form CO and CO_2 . Temperature increases also enhance polymer removal rates because of the Arrhenius dependence of the rate coefficient. Finally, ion, electron, and photon bombardment of the polymer breaks chemical bonds in large part via random chain scission [7] and promotes oxidant adsorption, polymer etching, and volatile product desorption [8].

Polymer structure and chemistry affect the oxidation and thus O₂-based etch rates of resist materials [9,10]. For instance, polymers with aromatic groups are more etch resistant than aliphatic materials. Polymers containing halogens display increased etch rates because of the fact that halogens are released as the polymer is degraded; the resulting F or Cl species generated in the plasma efficiently abstract H from the polymer chain, thus facilitating reaction with and removal of the polymer by O species.

Resist removal requirements initially were met by using parallel plate or barrel reactors at higher pressures than those used for etching, so that the reactant concentration was high and the ion energy was low [11]. Because radiation damage is a severe problem, downstream stripping configurations are dominant, using either MW or inductively coupled plasma (ICP) coils to create the plasma. In this mode, ion/electron/photon bombardment can be virtually eliminated, while establishing a high neutral species concentration. Plasma parameters (pressure, frequency, and power) can then be optimized to allow efficient generation of atoms and radicals and rapid transport (short residence time) to the resist surface [12]. A typical high-temperature, O₂-based, single-step bulk photoresist stripping process is shown in Table 7.2-1.

Achievement of high stripping rates at reasonable temperature (<200°C) is sometimes attained by adding selected gases or vapors to the O₂ plasma; note the additives in Table 7.2-1. As mentioned above, halogen atoms in the plasma atmosphere are effective in promoting resist removal because of H abstraction. In addition, it has been suggested that the presence of halogens, especially F,

TABLE 7.2-1 Typical High-Temperature O₂-Based Single-Step Photoresist Strip Process for Downstream Microwave (MW) or Inductively Coupled Plasma Process Modules

Bulk Photoresist Strip Process	Step 1
Gas flow	1000–4000 sccm O ₂
Additives	CF ₄ , H ₂ /N ₂ , N ₂ , H ₂ O
Process time	30–60 s
Strip rate	>1 μm
MW plasma	1000–2500 W
Radio frequency bias plasma	Off
Pressure	0.7–2.0 Torr
Temperature	200–350°C
Reference	[12]

markedly increases O atom concentrations, thereby further enhancing the removal rate [13–17]. As a result, additions of F-containing molecules such as CF_4 , SF_6 , and NF_3 are common in downstream stripping O_2 -based systems. Care must be exercised, however, to ensure that the long-lived F atoms do not etch other films or substrates. In fact, F-containing gases are frequently added to O_2 -based plasmas to promote the removal of Si or SiO-based residues; this will be discussed in more detail in Section 7.3.3.1.

Etching rates of polymer materials display a maximum with F-containing feed gas composition (10–25 atomic% F-containing additive) in both parallel plate reactors and downstream configurations because of the trade-off between enhanced O atom generation at low F content and O atom dilution at high-F content [18,19]. In addition, polymer surface fluorination to a depth of 30–50 nm occurs [20,21]. Fluorination of the polymer surface stabilizes the polymer or resist material, thereby inhibiting etch rates [21,22].

Other additives to O_2 also affect the removal rate of bulk photoresist or organic materials in a downstream reactor. Similar to halogen additions, N_2 addition increases the O atom concentration in O_2 plasmas and thus the resist stripping rate; because the effective activation energy does not change relative to pure O_2 plasmas, the mechanism of resist removal appears to remain that of attack of the hydrocarbon material by O atoms [23]. Addition of H_2O vapor to an O_2 plasma is analogous to the addition of F-containing vapors; i.e., strip rates increase up to $\sim 15\%$ H_2O addition, and the effective activation energy decreases [24].

Mechanistically, H_2O addition increases atomic O atom concentration in the plasma, and it also dissociates to give OH, which can extract H from the polymer backbone in a manner similar to F. Dissociation of the O_2 molecule is catalyzed by the addition of additives. These additives not only can increase the stripping rate of photoresist, they can change the selectivity to the substrate or film material underlying the photoresist. Comparison of the etch rates and effective activation energies of various additives to O_2 plasmas is discussed in Ref. [24], where the similarity of F- and H_2O -containing plasmas is described; likewise, the effect of N_2 addition is apparent [24]. Note also that achievement of high strip rates is possible at low temperatures ($<180^\circ\text{C}$) with CF_4 and H_2O additions. The addition of H_2 to O_2 plasmas increases the stripping rate [24] similar to that of H_2O addition; at low concentrations of H_2 , $<30\%$, the stripping rate increases, as shown in Fig. 7.2-4. However, at high concentrations of H_2 , the stripping rate decreases substantially when using an MW downstream reactor [24].

Water additions to stripping plasmas appear to have a further advantage in that Na contamination is passivated or electrically inactivated according to flatband shifts in metal oxide semiconductor (MOS) capacitors after resist strip [24,25]. Apparently, Na remains on the underlying SiO_2 surface, but is electrically inactive, in that it is immobile in the SiO_2 . That goes against the common theory that Na is very mobile in SiO_2 . A study by Savas [26] shows

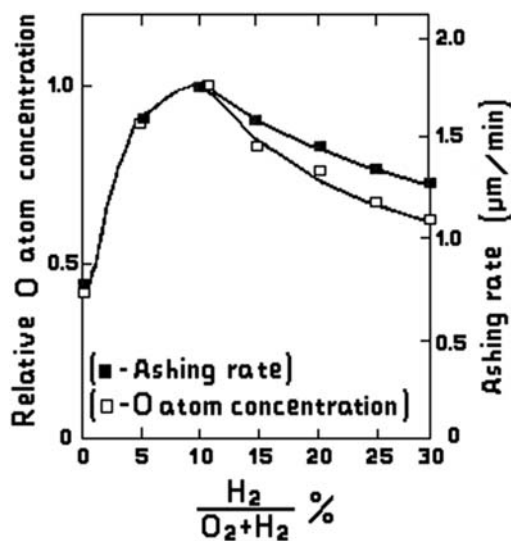


FIGURE 7.2-4 The stripping rate (ashing rate) of photoresist with respect to the atomic O concentration as a function of the $O_2:H_2$ ratio [24]. Reprinted with permission from S. Fujimura, K. Shinagawa, M.T. Suzuki, M. Nakamura, J. Vac. Sci Technol. B9 (2) (1991) 357. Copyright 1991, American Institute of Physics.

that Na, most likely originating in the photoresist, is driven into the SiO_2 layer by an increase in the overetching time of the plasma stripping step. When compared with the wet cleaning step, which should remove most of the Na, the 300% overetching sample shows approximately an order of magnitude increase in Na surface contamination, while the 100% overetched sample shows less increase in Na contamination, but never achieves the level of the wet cleaning process, as shown in Fig. 7.2-5.

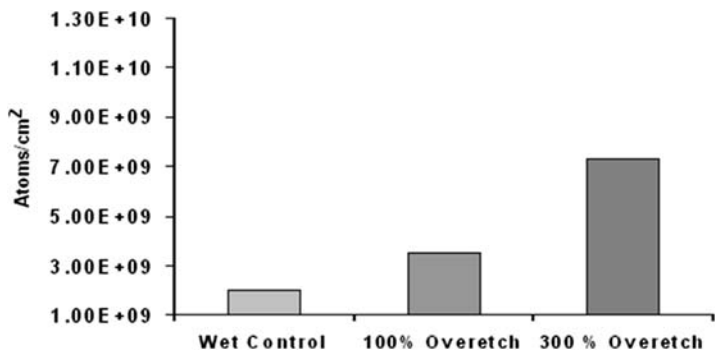


FIGURE 7.2-5 Surface Na contamination after plasma stripping processes with different overetch times compared with wet cleaning control wafer [26]. Used with permission from Solid State Technology.

7.2.3 Post-Ion Implant Resist Stripping and Cleaning

Post-ion implant resist stripping is difficult when high-ion doses are implanted and the photoresist is cross-linked and dehydrated during the process. High-dose implants are used when shallow source/drain junctions are formed, when the polysilicon gate material is doped, and when a collector implant is needed for bipolar-complementary MOS (BiCMOS) and bipolar devices. Fig. 7.2-6 shows the range of doses for implants used when a photoresist mask is appropriate [27]. High dose is usually defined as a dose greater than 5×10^{14} atom/cm² for As, P, B, or BF₂. Fig. 7.2-7 shows a schematic of the physical changes experienced by the photoresist during the high-dose implant. The crust that forms from the implant is highly cross-linked and dehydrated; also, the solvents that are uniformly distributed in the bulk photoresist have been outgassed in the top layer and the crust has less H than the unimplanted photoresist [28]. In addition, to the crust layer, the implanted species may be oxidized to As₂O₅, P₄O₁₀, and B₂O₃, none of which are volatile, and all of which are slightly hygroscopic. SiO₂ may be sputtered from the barrier layer and reside on the top or sidewall of the photoresist. All these residues must be removed to produce high-yielding devices.

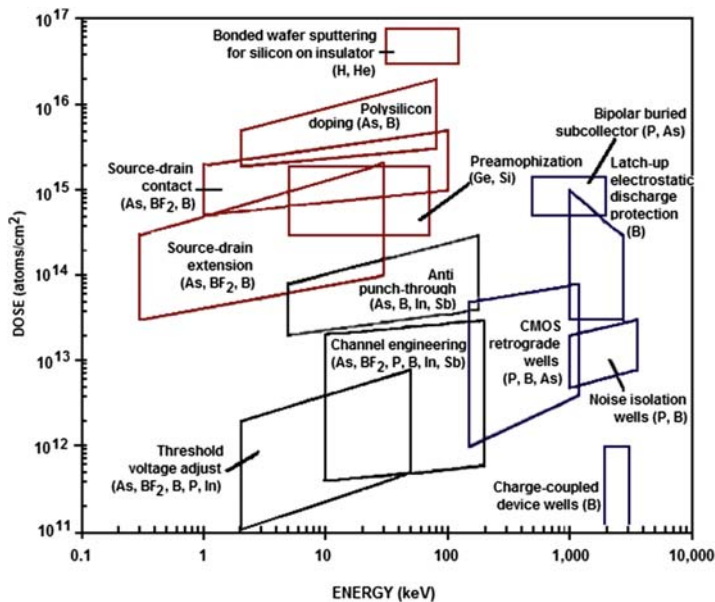


FIGURE 7.2-6 The high-dose implant regime is defined as greater than 5×10^{14} atoms/cm². Various implant layers are shown that use high doses [27]. Reprinted with permission from L. Rubin, J. Poate, *The Industrial Physicist*, June/July 2003, p. 12. Copyright 2003, American Institute of Physics.

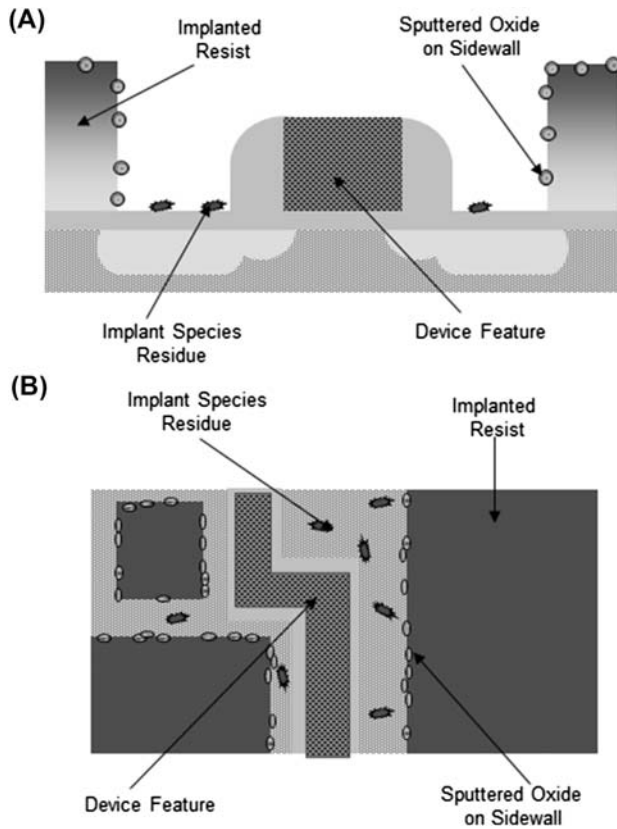


FIGURE 7.2-7 Cross-section view (A) top view of device after high-dose implant and (B) cross-section of the photoresist and features after a high-dose implant step.

A schematic of the cross-section of a high-dose implanted photoresist is shown in Fig. 7.2-8. The photoresist crust is modified compared to the underlying bulk photoresist, which is similar in chemical composition to the resist immediately after the photolithography process [29]. The crust layer is carbonaceous and does not behave like bulk resist during the stripping process [30]. This crust must be removed at a lower temperature to prevent the underlying bulk resist from “popping” [31]. Popping occurs when the bulk photoresist is heated and the entrapped solvent vaporizes; however, the crust inhibits the solvents from escaping and as the pressure builds up, popping occurs as illustrated in Fig. 7.2-8. The “popped” photoresist crust can redeposit on the wafer surface. This residue is extremely difficult to remove with a simple O_2 -based resist strip process and because of the nature of the contamination, which behaves similar to a particle, the wet cleaning chemistry must be tailored to remove this material. Sulfuric acid hydrogen peroxide

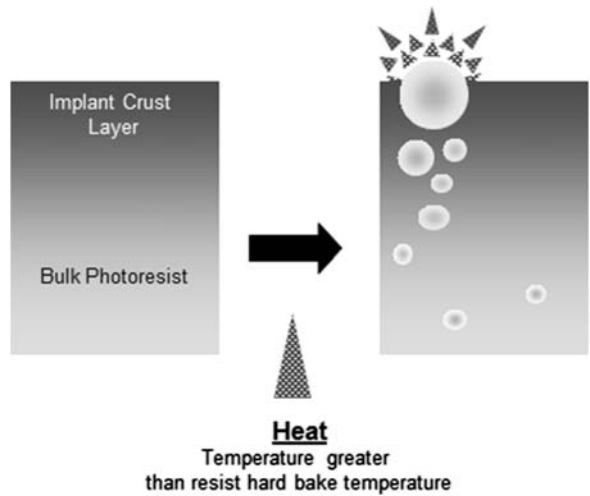


FIGURE 7.2-8 Schematic of a cross-section of high-dose implanted photoresist showing how “popping” occurs when the bulk photoresist is heated during the plasma strip step without first removing the crust layer.

mixture (H_2SO_4 and H_2O_2 , known as SPM), discussed in Chapter 4, does not always chemically attack this material [31].

Fujimura recognized as early as 1987 that high-dose implants photoresist stripping (HDIS) needed special process recipes to prevent excessive contamination from the popping of the resist [32]. Typical plasma stripping and cleaning processes for bulk photoresist stripping are shown in Table 7.2-1; while Table 7.2-2 shows a high-dose implant strip process. The bulk photoresist strip process is a simple one-step recipe, while the high-dose implant

TABLE 7.2-2 Typical High-Dose Implant Photoresist Strip and Clean Processes for Downstream or Inductively Coupled Modules			
Process A:	Step 1	Step 2	Step 3
Three-Step	Crust Removal	Bulk Photoresist	Final Photoresist
High-Dose Implants Photoresist Stripping (HDIS) Process		Strip	Strip
Gas flow	40–150 sccm O_2	1000 sccm O_2	1000 sccm O_2
Additives	H_2/N_2 , H_2O	None	None
Process time	30 s	10–15 s	<30 s

TABLE 7.2-2 Typical High-Dose Implant Photoresist Strip and Clean Processes for Downstream or Inductively Coupled Modules—cont'd

MW plasma	2000 W	1000 W	800 W
Radio frequency (RF) bias plasma	75 W	Off	Off
Pressure	5 m Torr	1.0 Torr	1.1 Torr
Temperature	40°C	<150°C	150°C
Process B:	Step 1	Step 2	Step 3
Three-Step	Crust Removal	Bulk Photoresist	Final Photoresist
HDIS Process		Strip	Strip
Gas flow	1000 sccm O ₂	2000 sccm O ₂	1000 sccm O ₂
Additives	None	None	20–50 sccm CF ₄ and 400 sccm H ₂ /N ₂
Process time	30 s	To endpoint	<30 s
MW plasma	1000 W	1000 W	800 W
RF bias plasma	100 W	Off	Off
Pressure	200–400 mTorr	1.3 Torr	1.3 Torr
Temperature	<90°C	>200°C	130°C
Process C:	Step 1a	Step 1b	
HDIS Crust Removal	Crust Removal 1	Crust Removal 2	
Process for 5×10^{15} atoms/cm² P			
Gas flow	67 sccm H ₂ O	67 sccm H ₂ O	
Gas flow	400 sccm He	400 sccm He	
Additive	None	3 sccm CF ₄	
Process time	40 s	20 s	
MW plasma	None	None	
RF bias plasma	500 W	500 W	
Pressure	650 mTorr	650 mTorr	
Temperature	15–140°C	15–140°C	

RF bias is typically used for the crust removal step coupled with the remote plasma. Process A [33], Process B [31], and Process C [34].

crust removal, photoresist strip, and residue removal steps all require different plasma parameters. Process A is a three-step process (1) crust removal, (2) intermediate bulk photoresist strip, and (3) final bulk photoresist strip. Process B adds a post-strip clean step using an F-containing additive. The F-containing gas effectively removes the implant residue and the sputtered oxide. The forming gas (H_2/N_2) treatment reduces the implant residue to volatile hydrides such as AsH_3 , PH_3 , or B_2H_6 . A DIW rinse or SPM process is suggested after the plasma process to ensure that all nonvolatile implant species have been removed.

The residue that may be present on an inadequately cleaned wafer is shown in Fig. 7.2-9. The typical residue is (1) “popping” residue, unreacted photoresist, (2) post-implant species residue, and (3) sputtered SiO_2 residue. Using the appropriate plasma strip and clean process is important, to target the removal of these residues or to prevent the residue from occurring, such as with the popped photoresist. The SiO_2 residue can be removed with the use of an F-containing gas such as CF_4 , NF_3 , or SF_6 . However, care must be taken to prevent uncontrolled etching of the SiO_2 film. Use of these gases is not

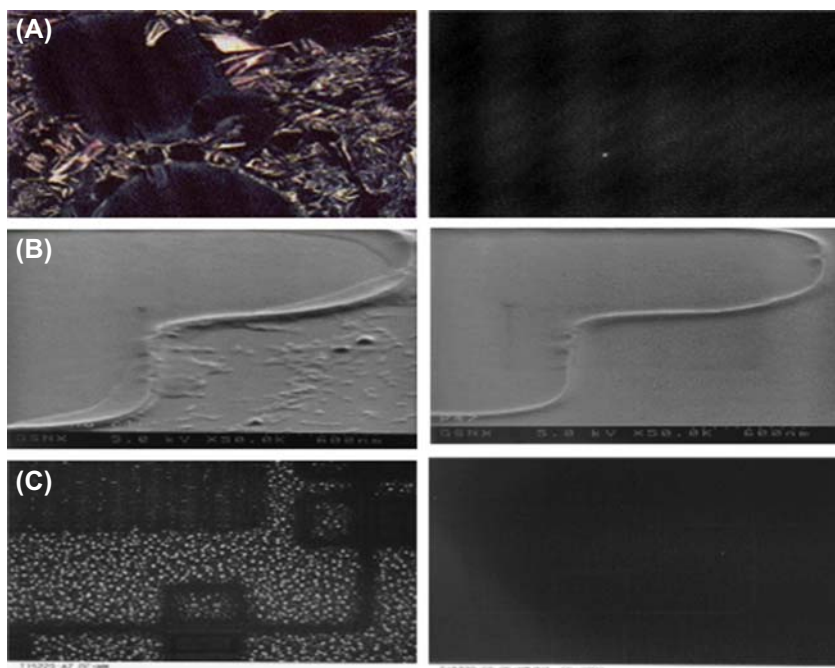


FIGURE 7.2-9 Micrographs of typical high-dose implant photoresist strip and clean process residues. The residue before clean is shown on the left and the adequately cleaned example is shown on the right. Examples are (A) “popping” of photoresist, (B) SiO_2 residue caused by implant sputtering of the barrier oxide, and (C) implant species residue. *Used with permission from Novellus Systems.*

suggested if the SiO_2 film is extremely thin (<2 nm), as there is a chance of breaking through the SiO_2 and etching the silicon. Pitting of the Si can occur if the etch rate of the Si is greater than that of the oxide. For these thin oxide films, further oxidation may occur, even with reducing chemistries [35].

Besides reducing the implanted species to volatile gases, forming gas (H_2/N_2) may also be used as an alternate gas to O_2 to achieve a slow photoresist etch rate and may be used in the crust removal step. Some crust removal steps use H_2/N_2 without O_2 . Also, H_2O vapor has shown some success for breaking through the implanted crust and accelerating the slow strip rate by using RF bias only. Helium or other inert gases can also be added. The bulk photoresist is then removed with a downstream O_2 -based plasma followed by SPM [34]. In most instances, the carbonized crust layer is removed by low-power RF reactive ion etching (RIE), either alone or in combination with MW power [34]. Fig. 7.2-10 shows partial stripping results using different approaches, such as stripping chemistry and intensity of ion bombardment, for post-implant stripping with crust removal [36]. Different chemistries result in different removal rates of the crust on top of the resist compared with the crust on the sidewall or the bulk resist beneath the crust.

Low-temperature crust removal, typically at a temperature lower than the photoresist hard-bake post-develop temperature, is used during the crust removal step to prevent the photoresist from “popping.” The bulk strip step

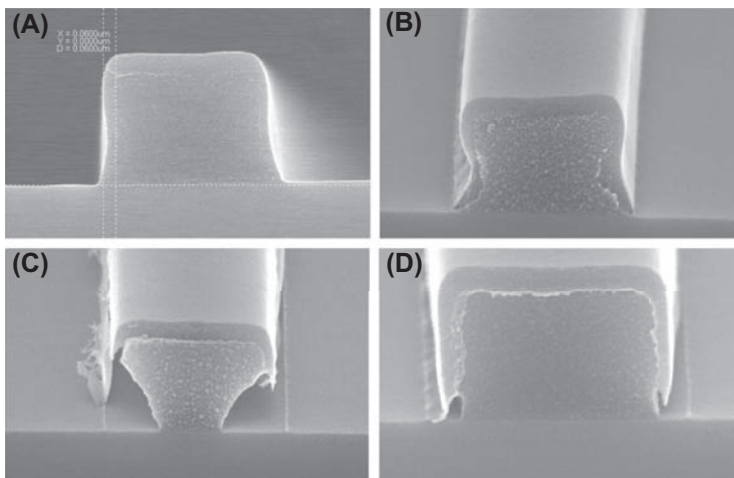


FIGURE 7.2-10 Cross-section of post-ion implant resist strip with different stripping rates of the top crust compared with the sidewall crust; (A) pre-stripping, (B) post-stripping with a ratio of the top:side = 0.28:1, where the sidewall crust is removed much faster than the top resulting in the bulk resist being etched under the crust, (C) post-stripping with a ratio of the top:side = 2.73:1, where the crust on top is removed much faster than the sidewall, and (D) post-stripping with a ratio of the top:side = 0.64:1, the preferable stripping rate ratio where the top and sidewall crust is etched at approximately the same rate [36]. *Permission.*

then can revert to the higher temperature to achieve a fast strip rate. The final clean step may also be performed at a lower temperature if an F-containing gas is used [37]. Direct plasma exposure as the first step with ion bombardment at room temperature is also effective to anisotropically remove the crust, which is followed by an isotropic bulk resist removal. This approach is effective in reducing the residue and defects and thus enhancing product yield [34]. Fig. 7.2-11 shows the comparison of different crust removal using direct ion bombardment with H₂- or O₂-based stripping chemistries.

A combination of dry and wet processes has proven successful in rapid stripping of high-dosed ion-implanted photoresist layers [38]. An atmospheric pressure inductively coupled hydrogen plasma was used to remove the crust layer followed by two treatments: sulfuric acid/ozone mixture and an ammonia/hydrogen peroxide mixture.

State-of-the-art ICs invoke a number of dielectric and semiconducting layers (e.g., silicon oxides, hafnium oxides, germanium, silicon) in FEOL

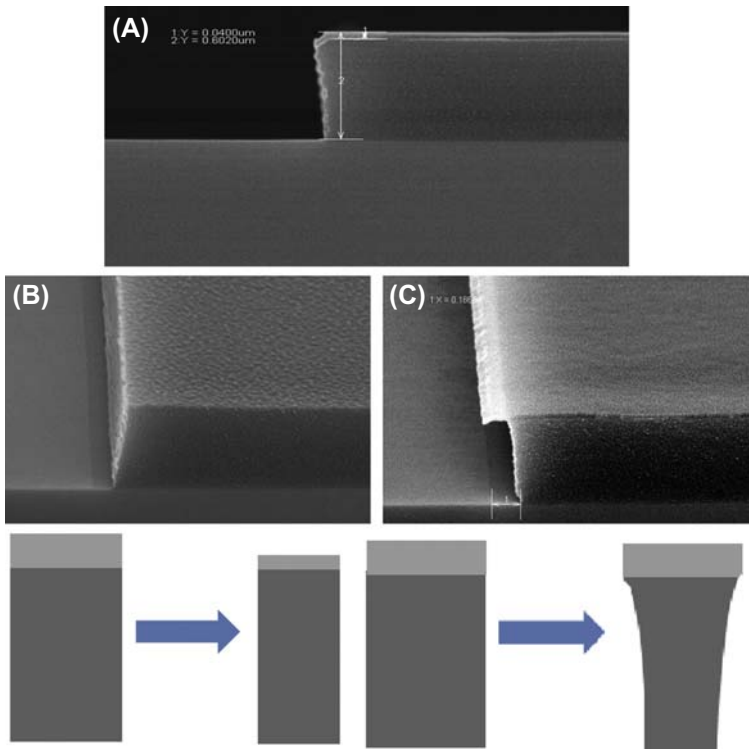


FIGURE 7.2-11 Cross-section of post-ion implant resist strip with ion bombardment; (A) pre-stripping, (B) H₂-based stripping chemistry that results in directional crust removal with more removal on the top, and (C) O₂-based stripping chemistry, that provides more sidewall bulk photoresist removal [36]. *Permission.*

processes, which present challenges for photoresist and especially HDI photoresist removal without damaging underlying layers. Removal problems are exacerbated as smaller device features, shallow junctions, new photoresists, and higher implant doses are employed [39]. Advanced wet chemical formulations have been used to remove HDI photoresists after a short O₂ plasma exposure; these results showed effective stripping and good compatibility with a variety of materials used in IC processes [40]. Resist stripping after the use of plasma doping as an alternative to ion implantation has shown promising results [41,42].

7.2.4 Post-etch Resist Stripping and Cleaning

7.2.4.1 Removal of Polymers

Polymer formation occurs at many steps in the manufacturing of IC devices. Some plasma etch processes purposely deposit polymer to protect the sidewall etching that erodes the critical dimension (CD). High-energy and high-dose ion implantation and high-density and energy plasma etch processes may cause cross-linking of the photoresist or dehydrogenation thereby creating a carbonaceous film that is extremely difficult to remove. In addition, sputtered residues can be embedded into the polymer, i.e., metals such as Al and Cu, or Si-based materials. Such materials further complicate removal of the polymer and associated residues. Nevertheless, this residue must be removed to ensure that the entire manufacturing process provides adequate electrical and mechanical performance. This may require that cleaning procedures remove impurities on surfaces to the sub-ppm level. Fig. 7.2-12 shows examples of post-etch residues.

Residues resulting from the etch process have been analyzed by various techniques. See Chapter 12 for review of the analytical techniques presented in this section. A few examples of sidewall residue analysis will be presented in this chapter. Although the residue has been defined specifically for these etch processes, other etch processes may lead to residues with slightly different chemical and physical properties and thus the stripping and cleaning process must be tailored to remove the residue from those etch processes. Optimization of a cleaning process involves various parameters, such as gas flows, power, and pressure; the following reference demonstrates the steps it takes to optimize a cleaning process [43].

Although plasma stripping can effectively remove photoresist, removing residue can be difficult. The difficulties arise when the residue is inert to the plasma and additives such as F-containing molecules must be added to remove the fluorocarbon polymer or the silicon sidewall polymer. The F-containing gases not only remove the undesirable polymer and SiO-like residue but also can undesirably etch the features and etch the underlying barrier or etch stop film. Metal residues are particularly difficult to remove with plasma alone. The metal residues may oxidize during the resist strip step and form

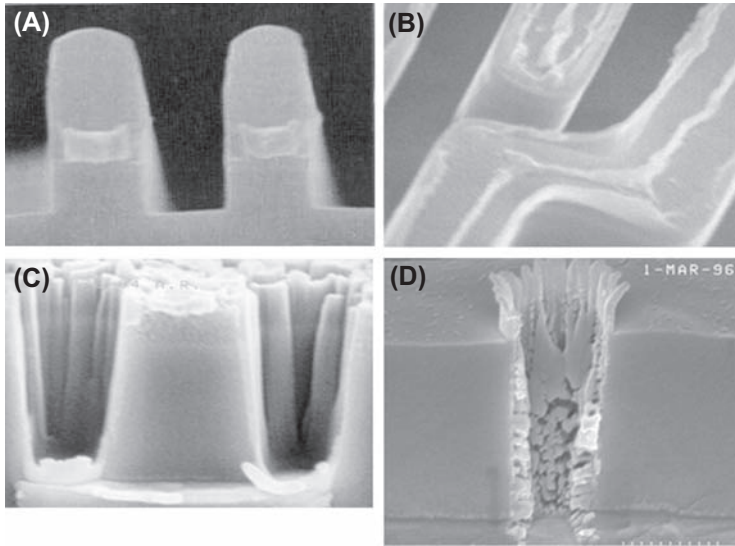


FIGURE 7.2-12 Examples of post-etch residue after etching and before stripping and cleaning, (A) and (C), and after etching and stripping but before cleaning, (B) and (D); (A) post-polysilicon gate feature after etching and before stripping and cleaning with photoresist, sidewall etch residue, and BARC, (B) post-polysilicon gate etch feature after stripping and cleaning with photoresist and BARC removed, with sidewall etch residue remaining, (C) post-contact etch feature after etching and before stripping and cleaning with photoresist, sidewall etch residue, and bottom etch residue, and (D) post-via etch feature after stripping and before cleaning with photoresist removed, with sidewall etch residue. *Used with permission of Sematech, Novellus Systems, and DuPont/EKC.*

nonvolatile and/or insoluble reaction products. Sometimes a reducing chemistry is used to prevent oxidation of the residues. However, not all metal hydrides are volatile; at times, a post-strip wet cleaning process is the only method that will remove the metal residue. In some cases, the plasma residue removal step can activate or alter residues to enable more effective removal by a subsequent wet cleaning step.

7.2.4.1.1 Cleaning Post-etch Polysilicon Gate Polymer

The gate electrode is one of the most important, if not the most important feature, of the MOS field effect transistor (MOSFET) and FinFET transistor. The gate electrode may be composed of polysilicon, a silicide material, and metal composites, such as WN, TiN, or TaN. The integration scheme for the formation of the gate structure depends on the device type; logic and memory devices have very different gate structures. The photoresist can be removed from the feature directly after the etch process or when a hard mask is used, the photoresist is sometimes removed after the hardmask etch, then the gate layer is delineated without photoresist, as the photoresist may adversely affect the etch process. These two MOSFET integration schemes are shown in

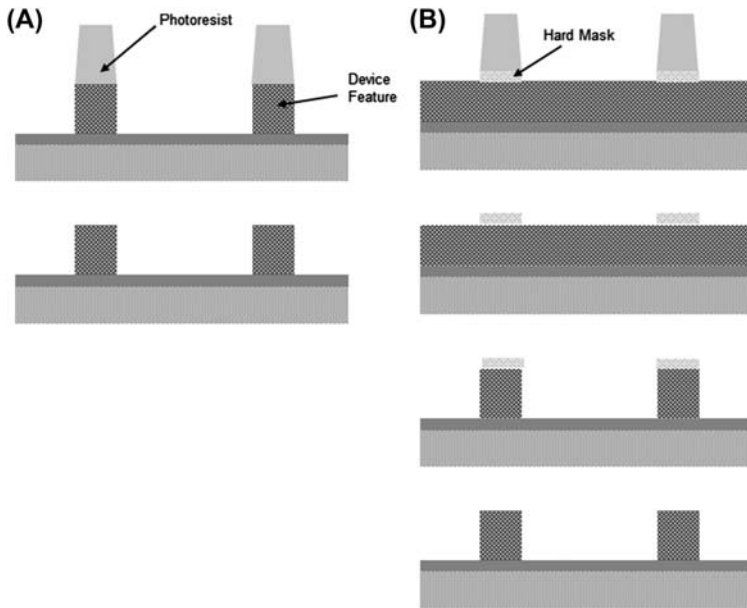


FIGURE 7.2-13 Various integration schemes for gate delineation (A) shows the gate structure is delineated with a photoresist mask and (B) shows the gate structure is delineated with a hard mask. The photoresist is first removed before the gate structure is etched.

Fig. 7.2-13. The hardmask may be kept in place, for example, if it is composed of SiO or SiN or may be removed if it is TiN or an etchable material such as amorphous carbon. Amorphous C may be simply removed with an O₂ plasma similar to that used in stripping bulk photoresist.

When TiN is exposed to O₂ plasma stripping chemistry, oxidation of this surface occurs. The researchers showed that even with pure reducing process chemistry, X-ray photoelectron spectroscopy (XPS) showed that both TiO₂ and TiN_xO_y are present as the top surface layers, whereas H₂/N₂ mixture had more TiO₂ formed [44].

The gate feature is typically etched with Cl- or Br-based chemistries, sometimes containing C, to provide directionality to the etching process by preventing lateral etching and thus a reduction in CD by passivation of the sidewall. Cleaning the sidewall polymer of the polysilicon gate feature or stacked silicide gate feature can prove difficult because of the resulting residues generated by the use of these gases. Fluorine-containing gases are also used to provide acceptable etch rates and oxygen-containing gases or O₂ are used to improve selectivity to the underlying SiO₂. Concentrations of 2–3 vol% of these additives have been successfully used to clean the post-etch residue on a dual-powered reactor as described in Section 7.4.1.1 [45].

Carbon is incorporated into the film either from photoresist decomposition during the etching process or from the etching chemistry [46]. Fig. 7.2-12A

shows a cross-sectional scanning electron microscope (SEM) image of a polysilicon gate after etching but before stripping and cleaning. The polymeric residue on the sidewall is clearly delineated as is the bottom antireflective coating (BARC) and the photoresist. Fig. 7.2-12B shows the top view of the polySi gate after strip but before cleaning; the residue “fence” is clearly visible.

The polymers formed on the sidewall are difficult to define quantitatively because of the small analysis area; the line height may only be a few hundred nanometers. However, some studies have reported detailed chemical information concerning such residues; see for instance [46–48]. Fig. 7.2-14 shows XPS spectra of sidewall polymer deposited during a polysilicon gate etch

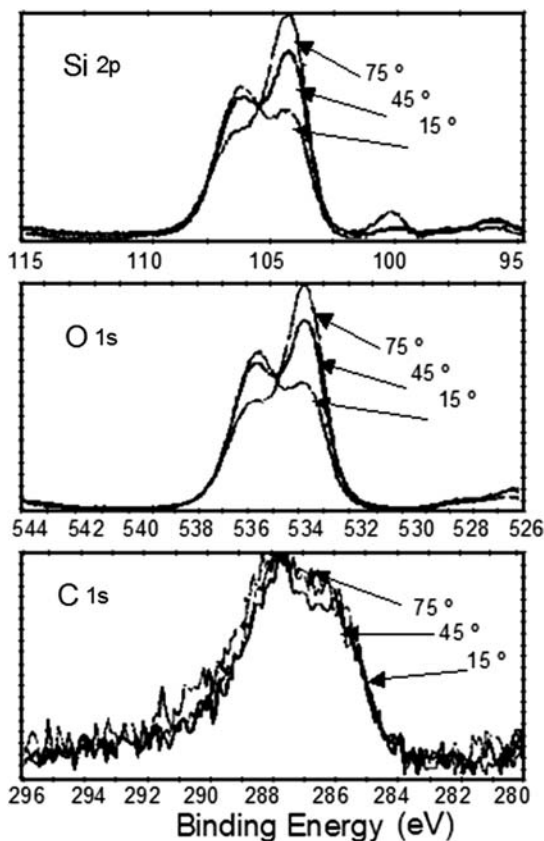


FIGURE 7.2-14 X-ray photoelectron spectroscopy spectra of post-polysilicon gate etch residue showing the bonding peaks for Si, O, and C [46]. Reproduced by permission of C.W. Lee, in: J. Ruzyllo, R.E. Novak (Eds.), *Fourth International Symposium on Cleaning Technology in Semiconductor Device Manufacturing*, vol. 95-20, The Electrochemical Society, Pennington, NJ, 1995, 243.

using $\text{Cl}_2/\text{HBr}/\text{He}/\text{O}_2$. The major components of the residue are Si and O with some C. The spectra indicate that the film is mainly composed of Si, apparently from byproducts formed during the etch process that deposit along with the C-containing sidewall polymer, although both Si and SiO_2 are present. The C observed can also be attributed to the presence of photoresist [46]. Analysis of this polymer also shows the presence of halogens, both Cl and Br, used for etching with the gas chemistry $\text{Cl}_2/\text{HBr}/\text{O}_2$. The approximate concentrations of the polymeric film on the photoresist sidewall are C 10, Cl 29, Br 3, Si 30, and O 28 at% (atomic percent). The polymer is slightly different on the photoresist as compared with the polysilicon feature. When using an oxide hard mask instead of photoresist, the polymer contains no C [47]. Further studies have shown that a thick SiO_xBr film is formed not only on the sidewall but also on the bottom oxide of the gate structure [48,49].

The sidewall residue remains after the plasma strip process and must be removed to prevent unwanted contamination of the polysilicon or gate oxide and to allow adequate adhesion of the next deposited layer. Because the sidewall is mainly composed of Si and SiO_2 , etching the sidewall will also etch the gate dielectric and the polysilicon gate features. Therefore, removal of the polymeric sidewall residue requires a process that minimizes the unwanted etching. Plasma stripping and cleaning recipes that minimize the gate dielectric loss are shown in Table 7.2-3. These recipes are for both ICP and MW downstream sources [50]. The processes are multistep recipes for stripping and cleaning the photoresist and residue from a polysilicon/tungsten silicide self-aligned gate stack. The processes are developed not only to remove the photoresist but also to remove the polymeric residue without etching the sensitive gate oxide, less than 0.5 nm is removed using either process shown in Table 7.2-3. Note that no RF bias is used in either recipe.

Although wet cleaning after an O_2 plasma photoresist strip process is quite common, the use of a plasma cleaning process can minimize the gate oxide loss while removing the sidewall polymeric material. The sidewall material is typically removed along with the photoresist in an O_2 -based plasma with an F-containing additive. Kabansky [51] recognized that for the post-strip clean, the combination of MW and RF power is required to retain the sidewall profile of the gate stack. In this case, when an O_2 -based plasma is used to etch a W-WSi-polySi structure, the addition of CF_4 , even with RF, causes degradation of the gate stack profile [51].

Surface oxidation of tungsten during photoresist stripping can present serious CD changes in the gate electrode. Indeed, a remote O_2 ICP used to strip photoresist from tungsten surfaces at or below 300°C oxidized the tungsten surface [52]. The extent of tungsten oxidation to form WO_3 increased with increasing temperature, but diminished when H_2 was added to the plasma. Oxidation was eliminated when sufficient H_2 was added despite an O_2 -dominant plasma atmosphere.

TABLE 7.2-3 Process for Post-gate Photoresist Strip and Clean Processes for Inductively Coupled Plasma Module (A) and a Downstream Microwave (MW) Module (B) [51]

Process A: Post-gate Stripping and Cleaning Process	Step 1 Gas Stabilization	Step 2 Plasma Stabilization	Step 3 Photoresist Strip	Step 4 Residue Removal	Step 5 Chamber Purge
Gas flow	2000 sccm O ₂	2000 sccm O ₂	4000 sccm O ₂	3000 sccm O ₂	2000 sccm O ₂
Additives	None	200 sccm N ₂	200 sccm N ₂	30 sccm CF ₄	200 sccm N ₂
Process time	Stabilization	2 s	45 s	20 s	0 s
MW plasma	0 W	950 W	900 W	900 W	0 W
Pressure	1.5 Torr	1.1 Torr	1.1 Torr	1.1 Torr	1.5 Torr
Temperature	Ambient	Ambient	200–300°C	200–300°C	Ambient
Process B: Post-gate Stripping and Cleaning Process	Step 1 Photoresist Strip	Step 2 Residue Removal			
Gas flow	2000 sccm O ₂	2000 sccm O ₂			
Additives	400 sccm 5%–10% H ₂ /N ₂	400 sccm 5%–10% H ₂ /N ₂			
Process time	90 s	60 s			
MW plasma	900 W	1100 W			
Pressure	1.2 Torr	1.2 Torr			
Temperature	130°C	275°C			

Similar concerns exist for unwanted silicon oxidation as the gate oxide layer thickness is reduced for advanced devices. That is, photoresist removal after gate stack etch exposes the remaining gate oxide in the adjacent stack regions to an O_2 -based plasma, which can increase the oxide thickness. These limitations may require the use of a reducing chemistry (H_2/N_2) for photoresist stripping in future devices.

7.2.4.1.2 Cleaning Post-etch Contact, via, and Trench Polymer

The contact and via features provide a conduction path between the working transistor and the metal layers or between the metal layers in an IC device. The contact and via structures are typically high-aspect ratio features, etched into a dielectric, that are subsequently filled with metal. Typical dielectrics used for such purposes are tetraethylorthosilicate, boron phosphorus silicate glass, or undoped silicate glass; however, low-dielectric constant (k) materials are currently more prevalent for low-capacitance/high-resistance interconnects for logic devices. Because metals (e.g., TiN, Al, and Cu) may be present at the bottom of the via, metal sputtering onto the sidewalls during the etch process can occur. The metal is thus embedded within the polymeric residue, requiring that this additional contamination be effectively removed to prevent a possible leakage path between the vias. Additionally, organic BARC films (e.g., polyimide) are used to assist delineation of deep contact and via structures to prevent reflective notching. Therefore, these films must also be removed.

Fig. 7.2-12C and D shows cross-sectional SEMs of the polymeric residues on the sidewalls of a contact and via. The contact SEM shows the feature before the stripping and cleaning steps, and polymer is clearly visible on the sidewall and the bottom of the feature. The resist is reticulated and dehydrated and likely has a polymeric crust. The via SEM shows the feature after stripping, but before cleaning, where the sidewall polymer is apparent and a “via veil” is evident.

The XPS spectra of the polymer formed during a via etch process and the surface after a cleaning process are shown in Fig. 7.2-15. The polymer was formed on a blanket Si wafer that was etched in a RIE plasma reactor. Polymer composition is mainly C and F, with O and a slight amount of Si [53]. The fluorocarbon that forms has been referred to as Teflon by many etch engineers. The bonding structure of the polymer formed in a HDP and a transformer-coupled plasma (TCP) or a magnetically enhanced plasma reactor (MERIE) are shown in Fig. 7.2-16 [54].

The polymer that forms during the dielectric etch process is highly cross-linked as shown in the XPS spectrum in Fig. 7.2-15; clearly, the HDP polymer has a larger amount of C cross-linking than does the polymer from a TCP or MERIE plasma. Nevertheless, both polymers have a significant amount of F making them very hard to remove in an O_2 -only plasma. The plasma clean chemistry that works best for removing these polymers contains F additives

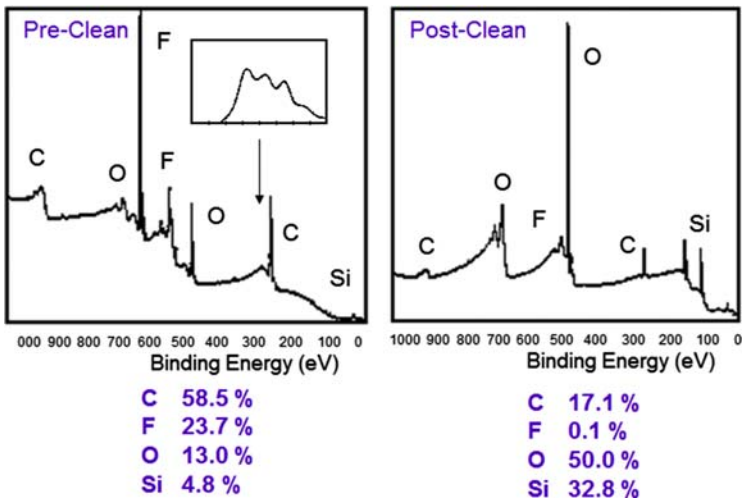


FIGURE 7.2-15 X-ray photoelectron spectroscopy survey spectra of dielectric etch residue showing the fluorocarbon-rich polymeric composition before and after cleaning the residue [53]. Reproduced by permission of P. Mikulan, S. Fonash, K. Reinhardt, T. Ta, in: J. Ruzyllo, R.E. Novak (Eds.), *Third International Symposium on Cleaning Technology in Semiconductor Device Manufacturing*, vol. 94-7, The Electrochemical Society, Pennington, NJ, 1994, 319.

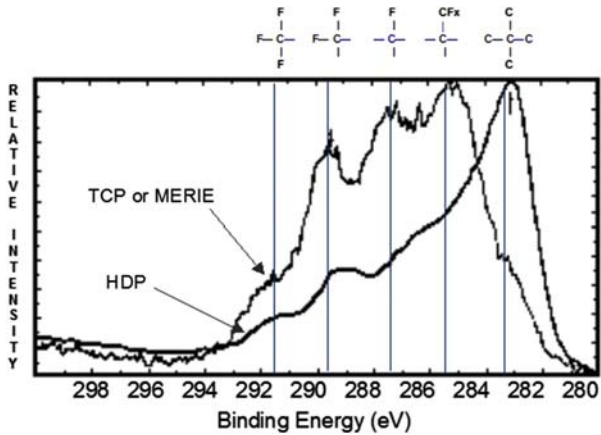


FIGURE 7.2-16 X-ray photoelectron spectroscopy high-resolution spectra of dielectric etch residue showing the C bonding structures for two different types of dielectric plasma etchers [54]. Reproduced by permission of M. Armacost, J. Marks, C.-L. Yang, in: G.S. Mathad, Y. Horiike (Eds.), *Proceeding of the Symposium on Highly Selective Dry Etching and Damage Control*, vol. 93-21, The Electrochemical Society, Pennington, NJ, 1993, 190.

such as CF_4 , NF_3 , and SF_6 , either with an O_2 -based or a H_2 -based plasma. The F-containing gases are able to etch the polymer, by forming volatile C_xF_y species, and can also etch any embedded Si in the polymer. However, because of the presence of F, the profile and CD of the device feature can be altered.

An additional component of the polymeric residue occurs when Si is present as an etch stop; an example is a dynamic random access memory contact etch. The C can react with Si and form a C—Si bond, as shown in the XPS spectrum in Fig. 7.2-16. As a result of the C incorporation, the C—Si layer increases the contact resistance. Furthermore, this C—Si bond is very stable, having approximately the same bonding energy as C—C bond. As a result, a typical O₂ plasma clean process does not easily remove this layer. The Si may also be damaged during the etch process and must be removed with a soft Si etch process. Fluorine may be added to the cleaning chemistry, which assists removal of the thin C—Si layer and the sidewall fluorocarbon polymer.

When metals such as Al or TiN are exposed to the plasma during the via etch process, metal may sputter onto the sidewall and combine with the fluorocarbon polymer. Fig. 7.2-17 shows an energy-dispersive X-ray spectrum of the sidewall polymer, the “via veils,” with embedded metal. In such cases, metal residues can be oxidized when using an O₂-based stripping chemistry (Fig. 7.2-18). Because aluminum oxide, Al₂O₃, is insoluble in most commercially available cleaning chemistries, either a reducing chemistry or F-addition is used to transform the Al oxide into a water-soluble product. Chapter 10 discusses surface chemistry of Al.

Many researchers have investigated the use of H₂-based plasma formulations for various applications such as cleaning Si before epitaxial deposition, as discussed in Section 7.2.6. More recently, to clean the polymer that is formed during the via etching process, interest has shifted from using O₂-based to H₂-based formulations. This change has occurred because of the ability to remove F-containing polymers in H₂-based plasmas, to prevent unwanted

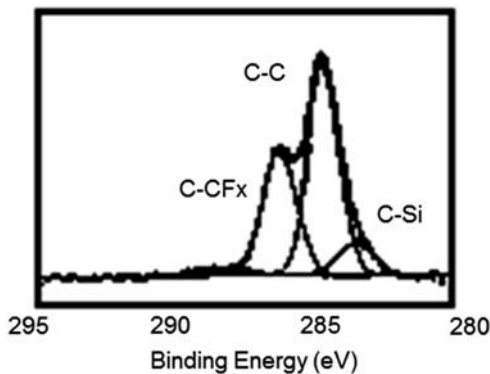


FIGURE 7.2-17 X-ray photoelectron spectroscopy high-resolution spectrum of dielectric etch residue showing the carbon-bonding structure in particular showing the Si-C bond—that is difficult to remove with conventional stripping plasma processes [55]. *Used with permission from Materials Research Society. Reprinted from K. Reinhardt, B. DiVincenzo, C.-L. Yang, P. Arleo, J. Marks, P. Mikulan, T. Gu, S. Fonash, An Effective In situ O₂ High Density Plasma Clean, Mat. Res. Soc. Symp. Proc. 315 (1993) 267, Pittsburg, PA.*

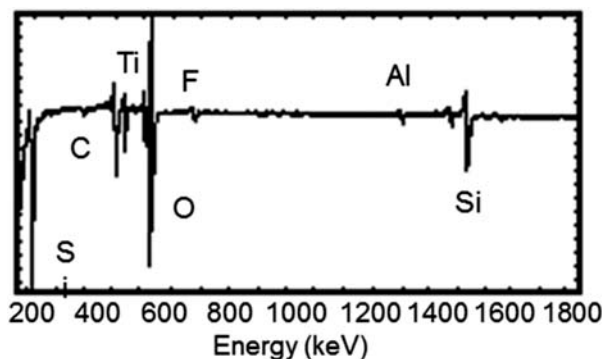


FIGURE 7.2-18 Energy-dispersive X-ray spectrum of dielectric etch residue showing sputtered metal embedded into the fluorocarbon polymer [56]. *Reproduced by permission of ECS—The Electrochemical Society from Y. Wang, S.W. Graham, L. Chan, S.T. Loong, J. Electrochem. Soc.*

oxidation of the barrier materials and also to prevent damage to the sensitive dielectric material. Somashekhar et al. have shown that H_2 -based plasmas in a downstream reactor are capable of removing polymer from via structures, both on the sidewall and the bottom without causing undesirable oxidation of the TiN over an Al metal layer [57]. Simko and Oehrlein have compared the cleaning ability of both O_2 and H_2 plasmas and the effect of these plasmas on the oxidation of Si [58]. Silicon oxidation by the O_2 plasma is very apparent, while minimal oxidation of the Si occurs in the H_2 plasma. However, O_2 is more effective at cleaning C from the surface, while H_2 is better at cleaning F from the surface, as shown in the XPS spectra in Fig. 7.2-19.

The dual damascene etching and stripping process flow is shown in Fig. 7.2-20, where the interlevel dielectric film is a low-k material. The cleaning method required when low-k dielectrics are present is dependent on the type of low-k material and the integration scheme used in the formation of the dual damascene structure. The dominant low-k film is carbon-doped oxide (CDO), an organosilicate glass. The C content and porosity of the low-k material are used to control the dielectric content. Increased C content imposes difficulties when using an O_2 -based plasma for stripping and cleaning because C is etched from the film, rendering the film SiO_2 -like. Inclusion of pores in the dielectric film causes difficulties in wet cleaning and drying that may require plasma cleaning to facilitate removal of liquid from the tiny pores, which in some cases are ~ 3 nm [59].

Stripping photoresist layers from the surface of ultralow k (ULK) nanoporous dielectrics represents a particularly difficult problem because of the existence of pores and the damage introduced into and oxidation of the ULK, thereby altering its electrical properties. H_2 -based stripping is therefore the method of choice. A capacitively coupled plasma reactor has been used to strip 193-nm and SiCOH-based photoresists [60]. Temperature increases from 200

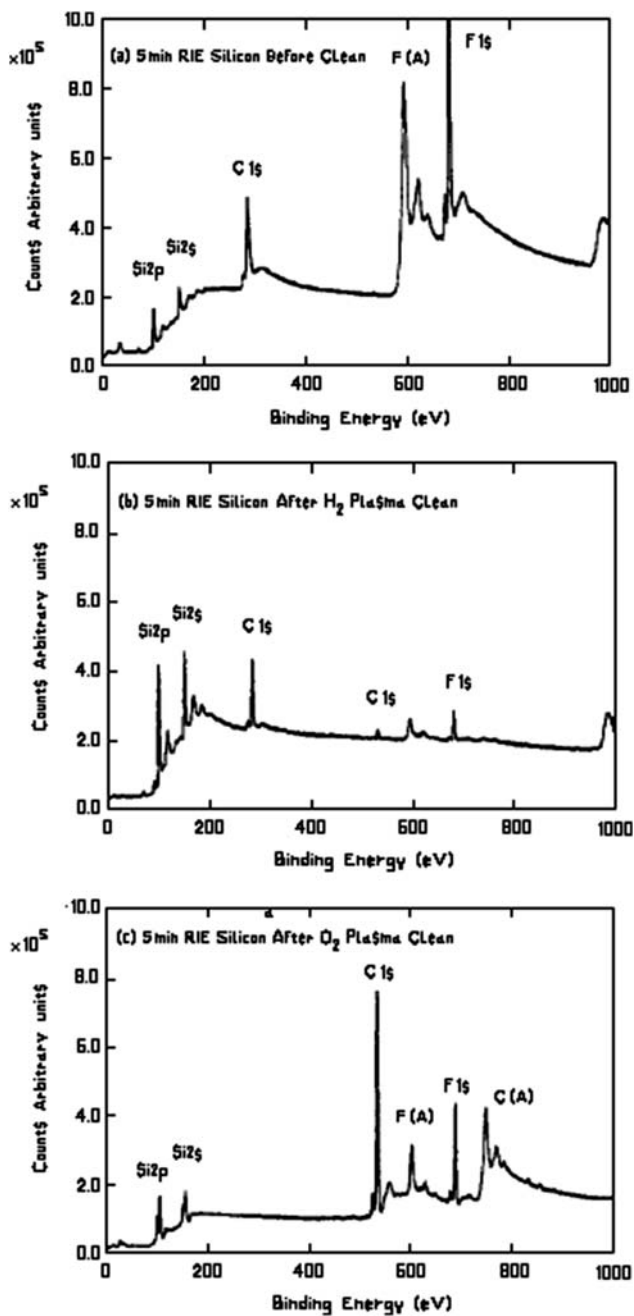


FIGURE 7.2-19 X-ray photoelectron spectroscopy spectra showing a comparison of O₂ and H₂ remote plasma cleaning of post-oxide etch residue [58]. Reproduced by permission of ECS—The Electrochemical Society from J.P. Simko, G.S. Oehrlein, T.M. Mayer, *J. Electrochem. Soc.*

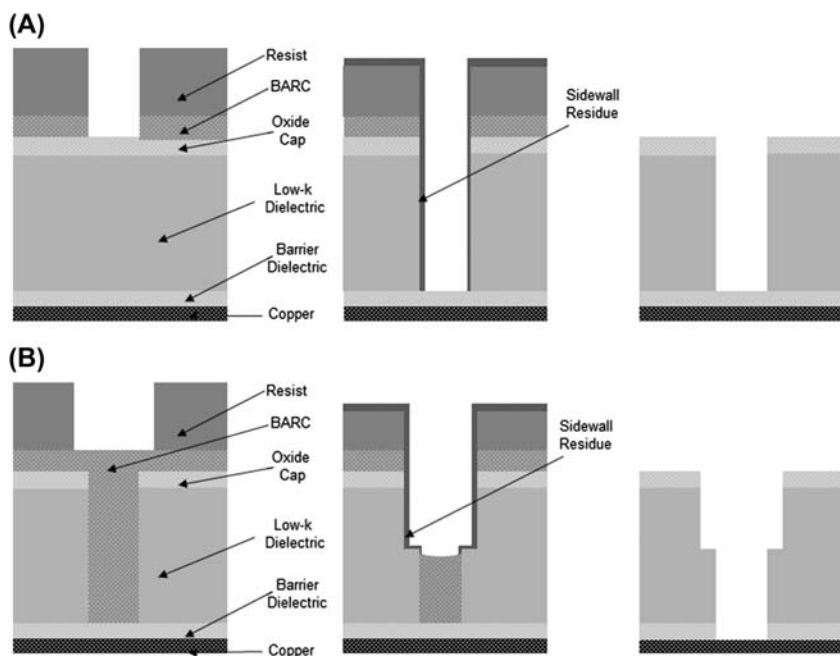


FIGURE 7.2-20 Cross-section schematic of dual damascene (A) via and (B) trench etch and strip process flow.

to 275°C increased the removal rate. N_2 additions to the H_2 plasma showed little effect on the stripping rate, but the damage depth to the ULK increased substantially as the N_2 content increased. After patterning ULK layers using a C_4F_8 plasma, most of the sidewall deposits could be removed using an H_2 plasma at 275°C.

Nanoporous organo-silicate low-k dielectrics typically show degradation in chemical and electrical properties because of the exposure to H_2 and H_2/N_2 plasma used for photoresist stripping [61]. High-carbon content ULK materials were especially susceptible to degradation as a result of the formation and volatilization of hydrocarbon fragments; such effects became more evident as the temperature was raised from 200 to 300°C. However, silica-rich organo-silicates showed minimal changes in chemical composition. Studies have shown that ions, as opposed to radicals, in the plasma are responsible for the majority of the methyl group depletion causing damage to the low-k material [62]. Additionally, vacuum UV (ultraviolet) photons produced by the plasma can penetrate the low-k materials, causing scission of the $-Si-CH_3$ bond, which accelerates the removal of $-CH_3$ groups [62].

Cleaning of the low-k dual-damascene structure requires successful removal of the photoresist and BARC films. No bowing or other modification of the via and trench profile, no change to the CD of the via and trench feature,

and no alteration of the dielectric constant from plasma damage may occur. The sensitive low-k material may be damaged during the cleaning process by abstraction of C from the CDO. Detrimental shrinkage of the film and an increase in the dielectric constant also occur. Contamination that may hinder a low-resistivity contact between metal layers, present at the bottom of the via, must be avoided. Sputtering of Cu during the photoresist stripping and residue removal process should be avoided, as this may lead to undesirable leakage and, in the worst case, a short between metal lines. Overviews of the challenges of low-k/Cu stripping and cleaning are outlined by Louis et al. [63] and Baklanov et al. [64].

Depending upon the specific process used, low-k films that are hydrophobic before the stripping and cleaning process may become hydrophilic and in some cases hygroscopic, thereby absorbing H_2O from the atmosphere and generating an adverse effect on the dielectric constant. Although baking or annealing sometimes eliminates this problem, water absorption is a serious concern with porous low-k materials [65]. Pore-sealing, in which the pores are chemically sealed by plasma exposure or other methods such as supercritical fluids, discussed in Chapter 8, is used to inhibit diffusion of H_2O or other unwanted impurities into the pores. However, plasma exposure often densifies the film and sidewall by forming a thin layer of SiO_x from the CDO material. Plasma recovery of the dielectric constant value after stripping and cleaning has been reported, by using a reducing [66] or by a CH_4 plasma [67]. Whether such processes recover the low-k properties by removing H_2O embedded in the material or by substitution of O in the CDO with H is unclear at present.

Stripping and cleaning polymeric films after dielectric etch is similar to cleaning polymeric films deposited during the polysilicon gate etch or to removal of the crust that forms during a high-dose implant step, described in Table 7.2-2. Both O_2 - and H_2 -based chemistries are used to strip the photoresist and to clean the sidewall and bottom polymeric residue. Oxygen-based chemistries are preferred because of the faster rate of photoresist removal; however, the O_2 -based cleaning recipes can cause severe damage to the C-rich low-k materials. In instances where O_2 plasmas may damage the low-k material, H_2 -based plasma recipes have shown promise. The addition of F can effectively etch the sidewall and bottom polymer. When the dielectric etch stop is a metal layer, the reducing chemistries and sometimes F-based chemistries may volatilize metal contamination, or at least render the metal soluble in H_2O -based cleaning chemistries. Oxygen containing chemistries oxidize the metal, creating nonvolatile products that may be insoluble in the subsequent wet cleaning solution. Water is sometimes used as an alternate plasma additive because it provides oxidizing capability, but is less aggressive than O_2 [68]. In addition, inert gases such as He or Ar may be added in place of N_2 .

Temperatures used to strip and clean low-k materials vary depending upon the specific process requirements. High temperature removes photoresist and residues rapidly and may deter the condensation of H_2O , while low

temperatures may limit the damage to low-k materials by reducing reaction rates with low-k films during the stripping and cleaning process. Ion bombardment controlled by substrate bias has been effective in removing residues on the sidewall and via bottom and the crust on top of the photoresist. However, ion bombardment can sputter exposed Cu. Typical processes used for low-k stripping and cleaning compiled from various sources are shown in Table 7.2-4. Cleaning chemistries used on the via layer that affect the subsequent trench photolithography step should be avoided [69]. Poisoning of the photoresist occurs when amines (bases) are present and the photoactive compound (PAC) is thus rendered inactive [70]. Table 7.2-4 shows that many plasma chemistries now exist for cleaning low-k materials; the wide range of gas chemistries, temperatures, and other process conditions suggest that there is no generally agreed upon or fundamental understanding currently available for the formulation of these recipes. Nevertheless, the process conditions reported have resulted in impressive results.

Removal of the photoresist after the etching process is sometimes performed in situ at the same time that the etch chamber is being cleaned of the built-up organic residue from the stripping process. However, transfer of the wafer to an ex situ plasma strip system is typically performed for most layers in IC manufacturing. Recipe LK7 is an in situ plasma-cleaning recipe. All other recipes in Table 7.2-4 are performed in separate chambers. In this case, the in situ strip process is performed to ensure simplicity, while the post-Al etch in situ strip is performed to meet a technical requirement: prevention of corrosion.

Savas et al. showed the results for stripping and cleaning photoresist and residue from low-k dielectric materials using an ICP reactor [75]. The researchers report that the strip process is efficient at stripping photoresist without removing or reacting significantly with the organic components of the low-k material because of the use of a low-pressure, O_2 -based, ion-activated process. Savas et al. suggest that the process is successful because it exposes the etched sidewalls to a higher ratio of ion flux than O radical flux and reduces the O diffusion into the low-k material; the process thus prevents abstraction of C from the sidewalls, while the ion impact effectively seals the sidewalls and inhibits O diffusion into the low-k dielectric [69]. Additional work on stripping low-k materials in an ICP chamber using various gas mixtures of N_2/O_2 , O_2/H_2 , and N_2/H_2 has been performed by Stucchi et al. [76], who reported that the gas chemistry selected affected the via profile.

As device dimensions and film thicknesses continue to decrease, etch selectivity to underlying films and substrates becomes increasingly important. Such concerns can be addressed by ensuring spatial uniformity of photoresist or residue removal, but this is difficult to control. Another approach is to establish high selectivity between photoresist/residues and other layers/materials. Specifically, the use of ALE or ALEt offers an alternative approach [77–80]. In ALE, as in atomic layer deposition, material is removed layer-by-layer,

TABLE 7.2-4 Processes for Low-k Dual Damascene Photoresist Crust Removal, Stripping, and Residue Removal Processes						
Low-k Stripping and Cleaning Process	Process					
	LK1	LK2	LK4	LK5	LK6	LK7
Process step	Crust removal	Crust removal	Photoresist strip	Photoresist strip/residue removal	Photoresist strip	Photoresist strip
Gas flows	3000 sccm O ₂ 150 sccm N ₂	80 sccm H ₂ 2000 sccm N ₂ 50 sccm CF ₄ option	5000 sccm 5% H ₂ /N ₂ 50 sccm CF ₄	3000 sccm H ₂ 150 sccm H ₂ O	3500 sccm O ₂ 3–15 vol% N ₂ option	550 sccm H ₂ 550 sccm N ₂
MW power	1100 W	1100 W	1500 W	1400 W	1400 W	1000 W (Top)
Radio frequency power	0 W	0 W	0 W	0 W	0 W	2000 W
Pressure	1.4 Torr	1.4 Torr	1.5 Torr	1.0 Torr	2.0 Torr	1.0 Torr
Temperature	25–250°C	Ambient	370°C	250°C	250°C	20–30°C
References	[71]	[68]	[72]	[73]	[72]	[74]

thereby offering unprecedented control of the process. Such an approach requires self-limiting reaction steps, e.g., either monolayer adsorption/reaction of an etchant species with the material to be removed or the ability to desorb/remove a monolayer of the product(s) formed. In many of these studies, reaction of the material with an etchant (with or without a plasma) is performed, followed by a low-power argon plasma exposure to desorb the product. Although most ALE investigations have considered Si removal, there have been other materials studied, including polystyrene [81]. In this study, the polystyrene surface was treated with O₂ to allow adsorption, residual O₂ was then evacuated from the chamber, and reaction products subsequently desorbed by a low-ion energy Ar plasma exposure. To avoid removal of polystyrene layers that were not modified by the O₂ treatment, a carbon source was needed to establish equilibrium between etching and deposition. A controllable etch rate of 0.1 nm per ALE cycle was achieved.

An alternative to residue removal in a plasma chamber is to modify the resist and residue, without stripping it completely, then using a wet cleaning formulation to remove the modified resist and residue to prevent damage to the sensitive dielectric material [82]. In this case, ion bombardment and vacuum UV is used to advantage to damage the polymeric residue, rendering it easier to remove in wet cleaning solutions.

7.2.5 Post-etch Aluminum Stripping and Cleaning

The post-etch residue after Al etch is unique because the polymer is purposely formed to prevent undercut of the lines during Cl₂-based etching. The Cl-rich polymer must be removed because if exposed to air, HCl will form, which will corrode the metal. Additionally, the photoresist collects the corrosive gases and vapors such as AlCl₃ and thereby contributes to the potential for corrosion because of out-gassing and makes stripping difficult because of the need to remove Al-containing residues [83]. Therefore, in situ stripping of the photoresist and cleaning of the residue are performed as the standard processes in most manufacturing facilities. Either partial or complete stripping of the photoresist and residue is performed in the etch chamber, typically followed immediately by wafer cleaning with solvents or H₂O-based cleaning formulations, to prevent corrosion. In this case, H₂O is not only used to strip remaining photoresist but also used to passivate the Al surface to form an oxidation layer to prevent corrosion. Typically, a plasma stripping chamber is installed on the same platform as a metal etch chamber to make sure the wafer is transferred under vacuum to prevent corrosion from exposure of chlorine-containing residues to the ambient moisture-laden atmosphere.

In some cases, as reported by Christaud et al. [84], after etching and stripping, wafers are transferred to the plasma cleaning system. The polymeric residue is removed using CF₄ and forming gas (N₂/4 vol% H₂) at a temperature of 75°C. Finally, the wafer is processed through a DIW rinse to remove

chloride-embedded surface residue. A short queue time between the etching and stripping is imperative to prevent corrosion caused by the formation of HCl from the reaction of the chloride ion with water vapor in the ambient atmosphere [85].

Because of the difficulty in stripping photoresist materials and residues after Al plasma etch, a sequence of steps is often used to facilitate removal with liquid cleans. For instance, a water rinse is sometimes used to remove chloride residues before a downstream O₂ plasma process. This approach can leave a particularly hard photoresist skin that is difficult to remove with subsequent liquid treatments as compared with a process without the water exposure [86]. Similarly, optimization of an O₂-based RIE plasma ashing sequence at low temperature (~15°C) instead of 275°C inhibits photoresist hardening and enables easier removal by solution stripping [87]. The common practice for post-metal etch is to use in situ water vapor strip, followed by a solvent clean, again a short queue time between the etch and cleaning steps.

7.2.6 Pre-epitaxial Deposition Cleaning

Epitaxial deposition is used to create an epitaxial (epi) layer (homoepitaxial or heteroepitaxial), such as Si, Ge, SiGe, SiC, or SiGeC on the Si substrate. The applications include epitaxial Si or SiGe on Si substrates for silicon on insulator, epitaxial wafer formation, and strained Si formation on the source drain areas for raised junctions. Remote plasma cleaning was first used for oxide removal from III–V materials and Ge surfaces [88]. In situ pre-epi deposition plasma-based cleaning processes have been developed for these applications [89]. In some cases, a single chamber can be used for both the cleaning cycle and the deposition cycle [90–93].

The goals of cleaning the substrate before epi deposition are the removal of native SiO₂ and adsorbed C, plus the removal of particles and metallic contamination, without roughening the surface [94]. Cleaning of such surface impurities is important for good adhesion and film resistivity control. Typically, pre-epi cleans are a two-step process (1) an RCA-type clean followed by (2) in situ or ex situ liquid or H-based vapor or plasma clean and passivation [95]. The RCA-type clean removes the particulate and metallic contamination along with the thick oxide. The in situ or ex situ clean removes the oxide and C contamination and passivates the surface with H to allow uniform epi deposition. Refer to Chapter 9 for discussion of the need for smooth and H-terminated Si surfaces.

There are many ways to effectively clean the Si surface before epi deposition. Chapter 6 discusses passivation of the Si surfaces using HF in the gas-phase and UV-assisted treatments. Another method uses a high-temperature H₂ bake (>900°C), which reacts with surface C and O to form volatile products. Typically, CVD epi deposition takes place at 700–800°C

and molecular beam epitaxy takes place at 600–650°C, although lower temperatures are desirable as discussed further in this section.

Yew and Reif have shown that physical sputtering processes, using Ar or Ar/H₂ plasmas, can be used to remove impurities from Si surfaces at low temperature; such processes require subsequent high-temperature anneals to recrystallize the surface, which is amorphized by the ion bombardment [96,97]. Because of this requirement, direct ion bombardment is rarely used for pre-epi surface conditioning. Another, less damaging method is remote plasma surface conditioning, using H₂, which is a viable low-temperature cleaning process for pre-epi deposition as discussed later in this section. The majority of remote plasma cleaning for pre-epitaxial and deposition has revolved around H-based plasmas.

The temperature should be low during epi deposition. The necessity for low-temperature epitaxial technology is because of the low thermal budget required to maintain doping profiles for shallow channel structures [98]. Plasma processes allow a higher reactivity at lower temperatures. For bulk epi deposition, it is not as important to have lower temperatures, as the dopants have not been introduced into the Si, except for BiCMOS and bipolar devices. Low-temperature remote H₂ plasma cleaning of Si surfaces for epitaxial growth at low temperature has been well documented [96,97,99], typically with a wet RCA-type preclean. In addition, electron cyclotron resonance (ECR) cleaning [100,101] has also been reported, and many CVD manufacturers have adopted this method of cleaning. Carter et al. has shown that cleaning Si in the presence of SiO₂ lines with remote H₂ plasmas did not adversely etch the SiO₂ lines or affect the roughness or cleanliness of the Si surface [102].

An RCA-type wet chemical treatment, typically performed before the H₂-based remote plasma cleaning, yields wafers with low levels of surface O and C. Streaks on the surface after epi deposition are often observed when the wafers are processed without H₂ passivation of the wafer surface, whether in situ or ex situ. In some cases, the streaks are formed from incomplete drying or from incomplete removal of metallic or organic contamination because of an unoptimized cleaning process. Refer to Chapter 4 for epi deposition wet pre-clean and Chapter 9 for the analysis of Si surfaces. Analysis of the epi surface is discussed by Montgomery et al. [103] and Delfino et al. [104,105]. Various analytical methods have been used to evaluate Si surfaces after cleaning and epi growth. For instance, XPS is a frequently employed technique; additional methods include spectroscopic ellipsometry for process optimization, discussed by Förster et al. [106], analysis of the properties of the strain in Si after H₂/Ar remote plasma cleaning in an ultra high-vacuum chamber are presented by Ramm et al. [107] and analysis of the surface by X-ray transmission electron microscopy is presented by Schneider et al. [108].

The efficiency of the remote plasma cleaning process is highly dependent on processing conditions; the plasma source and power, pressure, gas

chemistry and flow, process gas purity, and temperature. In typical remote plasma cleaning processes, atomic H etches the Si surface from room temperature to 700°C [109,110]. The etch rate decreases as the temperature is increased. These remote H₂ plasma treatments are very efficient in removing C-containing contaminants and also for removing native SiO₂ and passivating the Si surface [89,95,111,112]. Fig. 7.2-21 shows XPS spectra for a Si surface after cleaning with a remote H₂ plasma. Removal of O on the surface is critical for formation of a defect-free interface between the epitaxial layer and the substrate. The O on the Si surface can assume a variety of bonding states, and as the oxide thickness increases, the structure of the oxide changes [113]. Relatively small amounts of C and O that existed on the surface after wet chemical treatments are removed by the remote H₂ plasma treatment. Generally, it has been easier to remove C than O from the Si surface.

After the remote H₂ plasma clean process, the Si surface is structurally well ordered, free of O and C, and is H terminated. This allows low defect density deposition of both Si and SiGe, resulting in ultra-low temperature Si homo-epitaxy at temperatures as low as 150°C [114]. It is important to remove the H on the Si surface before deposition, which can be accomplished by annealing at a temperature greater than 500°C. Thomas et al. [111] showed that the temperature of cleaning is also important; at higher temperatures, 700°C,

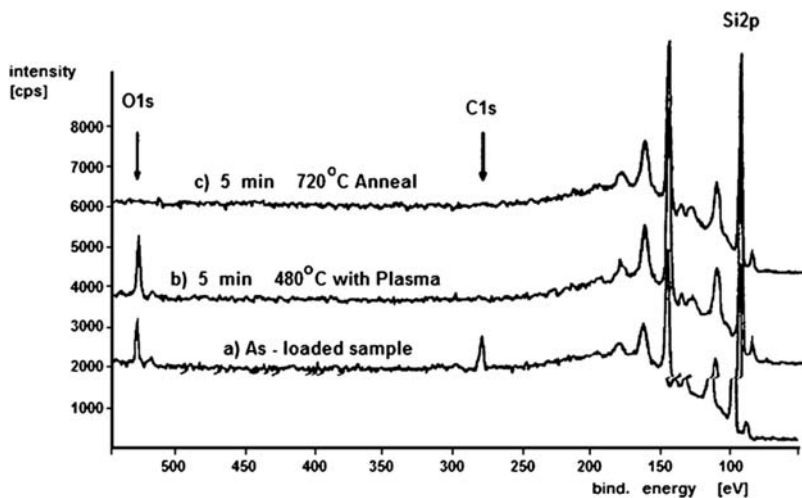


FIGURE 7.2-21 X-ray photoelectron spectroscopy spectra of a Si <100> surface after exposure to various processing conditions, (A) prior to processing, (B) after processing with remote H₂-based plasma, and (C) after anneal [111]. Reprinted with permission from R.E. Thomas, M.J. Mantini, R.A. Rudder, D.P. Malta, S.V. Hattangady, R.J. Markunas, *J. Vac. Sci. Technol. A*10 (4) (1992) 817. Copyright 1992, American Institute of Physics.

carbon was more effectively removed from a Si surface in H_2 plasmas than at 480°C . Kishimoto et al. [115] has shown that temperature also has an effect on the removal of SiO_x from the Si surface.

In situ processing is usually performed, either in the same chamber or in a separate chamber, on the same system platform, because a vacuum break may cause undesirable oxidation [116]. However, Strass et al. have shown that the Si—H terminated surface is stable for more than 24 h because of the presence of buried H in the Si crystal lattice when prepared with a H_2 plasma [116]. Passivation of the surface dangling bonds with H results in surfaces that are more resistant to oxidation than the non—H-terminated surfaces. Anthony et al. [114] showed that Si surfaces were more resistant to oxidation and demonstrated the effectiveness of H_2 plasma treatments for removing O from oxidized surfaces. The experiment showed that 10 h after a modified RCA treatment, substantial C and O appear on the surface of a Si wafer when exposed to atmosphere even though immediately after the wet cleaning step there was little O on the surface and no Auger electron spectroscopy (AES) detectable C. However, after the wet cleaning steps and exposure to a H_2 plasma, O was removed from the Si surface. Atmospheric exposure of the sample resulted in an uptake of O similar to what was observed after wet chemistry, but the surface showed a significantly lower uptake of C. Ramm et al. showed using SIMS depth profiles that at the Si epi—wafer interface there was approximately 1×10^{-3} monolayers of O and less than 1×10^{-3} monolayers of C when H_2/Ar plasma cleaning of the wafer surface was performed [117].

7.2.7 Pre-high-k Deposition and Oxidation Plasma Cleaning

Cleaning of the silicon surface before high-k deposition is usually performed with an HF vapor and UV clean. However, remote H_2 -based plasmas are also effective at removing unwanted oxides and preparing the surface for deposition in the same manner as pre-epitaxial deposition cleaning [118].

7.2.8 Pre-metal and Pre-dielectric Deposition Plasma Cleaning

Cleaning of features before metal deposition is critical, as native oxide and other contaminants may adversely affect the electrical properties of the IC device. Native oxide forms on the surface from exposure of Cu films to an O_2 -containing atmosphere; the resulting surface layer is composed of Cu_2O , CuO , $Cu(OH)_2$, and $CuCO_3$ [119]. This native oxide prevents a uniform conduction path between two metal layers, thereby increasing the resistance. Furthermore, contamination, especially metals, establishes a leakage path by diffusing into the dielectric. Additionally, an uneven thickness distribution of native oxide prevents uniform deposition of the metal, sometimes completely inhibiting film deposition. Chapter 10 discusses Cu surface morphology. To remove

native oxide on Cu, an in situ clean step is typically performed immediately before the barrier metal deposition step. This step can either be performed in the same chamber as the deposition or in a chamber adjacent to the deposition chamber, to avoid a break in the vacuum and thus exposure of the wafer to O₂. Sputtered metals from the preceding etch process or from the etch reaction and post-etch polymeric residues can also be present. Wet chemical cleaning processes usually precede the pre-deposition cleaning step. These wet cleaning processes are discussed in Chapter 4.

One method of pre-deposition cleaning is sputter etching. Sputter pre-cleaning of contact and via features is used when the underlying substrate material is relatively resistant to the plasma etch processes. Sputtering causes redeposition of Si and SiO₂, and sputters soft metal layers such as Al and Cu. Copper devices usually do not use sputter etching, but rely on downstream H-based cleaning. Sputter etching is used for Al stack deposition pre-cleaning and is effective for removing native oxides and contaminants in large features or in small features (<0.5 μm) with aspect ratios of the order of ~4:1. Care must be taken to prevent resputtering of material onto the sidewalls of features. For features having high aspect ratios, resputtering of material from the bottom to the sidewalls of the feature during the cleaning process may result in significant narrowing of the feature and a corresponding increase in resistance of the feature. Also, sputter etching may widen some features, cause a retrograde angle to the feature cross-section, or round the top corners, all of which may impair device performance.

A remote ECR plasma cleaning process before CVD of Ti uses a mixture of Ar and H₂ [120]. Argon is usually used in sputter etching processes and provides significant ion bombardment. Although this is advantageous for sputtering impurities from the bottom of the feature, the bombardment may resputter the underlying metal onto the sidewall. By using a remote plasma with Ar, directionality is provided and resputtering is minimized. The addition of H₂ allows removal of the native oxide and promotes formation of a uniform TiSi_x layer between the Si substrate and the deposited Ti.

Cleaning Cu with a H₂ downstream source is disclosed in US patent 6,905,965 [121]. First, an F-containing plasma is used to remove SiO₂ and fluorocarbon polymers, then an in situ reducing gas is used to convert the CuO to Cu without sputtering the Cu sublayer. Table 7.2-5 shows the recipe used for this process. This process is used before barrier deposition of Ta/TaN in a physical vapor deposition chamber or TiN in a CVD chamber. Characterization of the Cu surface after a thermal treatment and exposure to reducing plasma adequately removes the CuO_x layer [105]. In this process, H₂O and CO₂ desorb during the thermal treatment and the H₂ plasma reduced the remaining CuO_x at temperatures above 150°C.

Falkenstein et al. reports another method for cleaning oxides from Cu surfaces by using an Ar and fluorocarbon (C₂F₆) plasma treatment [122].

TABLE 7.2-5 Process for Removing Oxides From a Cu Surface Before Barrier Deposition [121]

Process Step	CuO Reduction
Gas flows	90% H ₂ /10% He, Ar or N ₂
Radio frequency (RF) 400 kHz power	300 W
RF 13.56 MHz power	10 W
Pressure	2–20 mTorr

However, stress migration of the Cu can be induced because of an aggressive Ar sputter etching pre-cleaning step [123].

7.2.9 Other Applications of and Approaches to Plasma Cleaning

Soft etching of a surface previously exposed to a plasma is another surface conditioning method. These processes remove the thin damaged layer with a slow, controllable etch rate that avoids resputtering of materials from the bottom of the feature onto the sidewall and prevents further damage. Soft etching is generally implemented after an etch process, specifically contact etch, where the underlying Si is damaged and thus would yield high contact resistivity; this process may be performed in situ or ex situ. The plasma system must be capable of providing a slow, controllable, and uniform etch, with adequate selectivity. In principle, any etching system may be used; however, plasma resist stripping systems, especially those configured with RF bias capabilities, are typically preferred. A soft etch process that uses an ICP with Ar to remove native oxides from a Si substrate has been described by Sumi et al. [124]. Alternatively, soft etching processes can use an F-based chemistry in situ within an oxide etching chamber [125]. In this process, inert gases such as Ar and He have been eliminated because they caused additional substrate damage. In this case, F-containing gases alone provided acceptable low etch rate and good uniformity.

Atmospheric pressure plasma jet cleaning is used for various industrial applications such as removing paint and cleaning metal surfaces of organic contamination [126,127]. Arc plasma jet cleaning of the Si surface before CoSi₂/Si contact formation has been reported by Ya et al. [128]. The plasma jet cleaning improves Schottky barrier contact parameters when compared with conventional aqueous HF cleaning and subsequent in situ Ar-ion-beam sputter etching. AES has been used to demonstrate the removal of O from the Si surface. In addition, longer exposure times between final cleaning and metal deposition may be allowed with no decrease in resistivity.

Medium pressure plasma jets have demonstrated the ability to remove resist materials [129]. Reactants (O_2/N_2) were activated in a surface wave discharge plasma at pressures between 50 and 80 Torr and directed onto the photoresist surface ~ 1 cm downstream. Removal rates of $2.5 \mu\text{m}/\text{min}$ were achieved.

Laser irradiation (Nd:YAG at a fundamental wavelength of 1064 nm with a second harmonic of 532 nm) has been used to remove diazonaphthoquinone resists from a silicon surface without damage to the silicon [130]. The process was performed under water. Resist removal was believed to occur as a result of pressure from the water and thermal expansion of the silicon from laser irradiation.

7.2.10 Cleaning and Conditioning Plasma Chambers

The plasma chambers used for etching, deposition, and cleaning can become coated with the byproducts of the various reactions performed. This deposited material can detrimentally affect the process by lowering removal rates or deposition rates and also create particles that may deposit on the wafer surface. In some cases, it is advantageous to coat the chamber wall with polymer or condition the chamber with gases that are used for the etching or deposition process. After the vacuum chamber is opened to the atmosphere, H_2O may permeate the chamber material and thus must be degassed before subsequent wafer processing. After a wet chamber clean, the cleaning agents may not be completely removed and they may adversely affect the process. Plasma strip chambers usually do not have the stringent requirements necessary for etch or deposition tools; nevertheless, these chambers do experience a build-up of photoresist byproducts because of unreacted organic material, especially when a H_2 -based process is used. Periodic cleaning of the chambers is advantageous to remove the unwanted residue and assure that particles are removed to avoid impingement onto the wafer surface.

Cleaning of a deposition chamber such as that used to deposit W is critical to eliminate the flaking and peeling of layers deposited on the walls of the chamber. Fluorine-based cleaning formulations are preferred, especially those that do not add fluorochlorocarbons or other ozone-depleting gases to the environment [131]. Discussion of this topic is beyond the scope of this book; however, many references exist for this cleaning method [3,132–135].

7.3 MECHANISMS OF PLASMA STRIPPING, CLEANING, AND SURFACE CONDITIONING

7.3.1 Overview of Plasma Parameters

Achievement of optimum plasma generation requires control and understanding of the important parameters that contribute to the creation of a stable

TABLE 7.3-6 Overview of Plasma Parameters for Stripping and Cleaning			
Pressure	Power	Gas Flow	Temperature
High	Plasma Sources	Gas Types	High
>1.0 Torr			>65°C
Low-energy ion bombardment, more chemical effects	MW downstream or inductively coupled plasma (ICP)	Oxidizing: O, O ₃ , N ₂ O	Prevents popping during high-dose implants photoresist stripping
Used for isotropic cleaning and stripping	Reactive ion source radio frequency (RF)	Reducing: H ₂ , NH ₃ , H ₂ O, H ₂ /N ₂	Prevents impurities from diffusing into oxide
Usually high flow of gas is needed to obtain high pressure	Reactive ion etching (RIE) + MW/ICP	Additives: Fluorine: CF ₄ , NF ₃ or others inert: He, Ar, N ₂	
Low	Power Ranges	Flow Ranges	Low
>100 mTorr			>100 mTorr
Higher energy ion bombardment, chemical and physical effects Used when RIE needs directionality	MW: 500–2500 W Reactive ion etching (RIE) + MW: 75–100 W RF RIE: 500–2000 W	Low pressure: 10–500 sccm High pressure: 500–2000 sccm	Used for rapid stripping of bulk photoresist and also for low-k cleaning

and repeatable plasma. The key parameters that control the plasma and contribute to the results of the process are outlined in [Table 7.3-6](#).

7.3.2 Overview of Physical and Chemical Mechanisms for Stripping and Cleaning

Remote plasma cleaning of surfaces during the manufacture of ICs has been successfully implemented for a number of technology generations. Remote plasmas may be essentially directional (anisotropic) or multidirectional (isotropic), defined by the parameters of power and pressure. The addition of a bias can increase the directionality of the etch process.

The physical and chemical mechanisms for removal of photoresist, impurities, and residues will be discussed in this section. The controlling

mechanisms for photoresist and impurity removal are important because undesirable effects such as surface roughening, damage of sensitive films, or incomplete removal of residues are likely when developing a process, and knowledge of the mechanism will aid in establishing a process around these challenges. Specifically, when developing a cleaning process, plasma reactions with the film to be removed, the surface contaminants, and the substrate must be carefully considered.

Plasma etching and stripping processes use both physical and chemical effects to remove photoresist, residues, and impurities from the wafer surface. Momentum transfer (sputtering) by ions physically dislodges residue and photoresist and breaks surface chemical bonds, and the reactive atoms and radicals chemically react with residue or photoresist to form volatile or soluble products. High-atomic weight atoms such as Ar, are especially useful for sputtering, whereas gases or vapors that dissociate to yield highly reactive atoms promote chemical reactions. Especially reactive species include F and O.

An O₂-based process easily strips hydrocarbon photoresists but is somewhat limited in removing residues such as fluorocarbons and metals. Another approach to chemical removal relies on reducing gases. An H₂-based process also strips hydrocarbon residue, but at a much slower rate than is obtained with an oxidizing process. However, H₂ can reduce some metallic elements to volatile hydrides. To remove the nonvolatile residues, a liquid cleaning process is usually needed after the plasma process.

7.3.3 Overview of Chemistries Used for Plasma Stripping and Cleaning

Many different chemical formulations are used to remove photoresist and organic contamination from wafers. Four general categories can be used to classify the gases: oxidizing, reducing, inert, and reacting. Table 7.3-7 shows the gases generally used for photoresist stripping, residue removal, and surface passivation.

TABLE 7.3-7 Categories of Gases Used for Strip and Clean Processes

Oxidizing	Reducing	Inert	Reacting
O ₂	H ₂	He	CF ₄
O ₃	H ₂ O	Ar	NF ₃
N ₂ O	NH ₃	N ₂	SF ₆
O ₂ /N ₂	H ₂ /N ₂		C ₂ F ₆

Byproducts are produced during the photoresist stripping process, some of which are volatile; however, some of the byproducts are incompletely reacted species that deposit on the walls and vacuum ports of the chamber. After a critical thickness of residue builds up on the chamber wall, the film starts to peel and flake from the surface, generating particulate contamination. Both these types of build-ups require regular maintenance. The residue formed may be unreacted fluorocarbon moieties or implanted species. Photoresists for extreme ultra-violet lithography contains metal oxide that will leave metal residue on the chamber wall. Furthermore, the currents flowing within the plasma chamber are affected by the composition and thickness of the chamber deposits, thereby undesirably influencing the reproducibility and control of the etch rates unless frequent cleaning is performed. The chemistry and the plasma source used for cleaning clearly have an effect on the amount and type of residue deposited.

7.3.3.1 *Chemistries for Photoresist and Residue Removal*

Various plasma stripping and cleaning chemistries currently exist for removal of photoresist and residues. This section will cover the most common chemistries used for stripping and cleaning. The chemistries used for photoresist stripping and cleaning are complex mixtures; gone are the days when a simple O_2 plasma was used for “ashing.” The chemistries used for stripping and cleaning devices with features 45 nm or smaller are usually composed of multiple gases, all at tightly controlled levels to prevent undesirable etching of the sensitive underlying oxide and Si films. Furthermore, the photoresist and residue are a complex mixture of hydrocarbons, halocarbons, carbon, embedded metallic atoms, etc. Also, the organic-based residue may contain F, making the film Teflon-like. Materials such as metallic contaminants that are sputtered from etching processes may be embedded in the polymer. The photoresist may reticulate during etching or other processes [135]. Many methods have been developed to remove the organic films. This chapter will focus on plasma methods. An overview of photoresist stripping from a lithography standpoint is given in reference [136].

The plasma process removes photoresist by reaction of plasma-generated species with the organic photoresist material. Photoresist removal is thought to occur through a process of hydrogenation or oxidation followed by volatilization. The photoresist can “burn” easily at high temperatures typically caused by decomposition of the higher molecular weight hydrocarbons and rapid vaporization of solvents. The photoresist tends to shrink and form a reticulated structure [137], as shown in Fig. 7.3-22. In some cases where photoresist is exposed to higher temperatures, a UV bake is performed to prevent reticulation. Reticulated photoresist is usually harder to remove than standard photoresist from temperature-assisted polymer cross-linking; as a

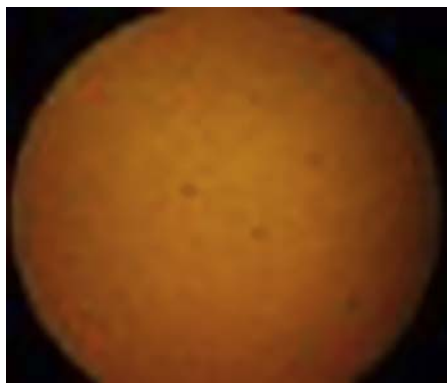


FIGURE 7.3-22 An example of reticulated photoresist [137]; clearly, the photoresist is wrinkled in appearance. *Used with permission from Matthew Last, Berkeley Sensor and Actuator Center (BSAC).*

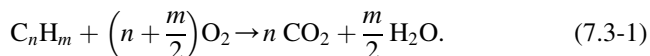
result, care is usually exercised during IC manufacturing to avoid this undesirable occurrence.

In addition to photoresist, other materials are used in the lithography process. Before photoresist is applied to the wafer, a priming agent such as hexamethyldisilazane is usually coated on the wafer to promote adhesion. Because of the Si contained in this primer, residues can remain after plasma resist stripping. Organic antireflective coatings, both below the photoresist (BARC) and above the photoresist, must also be removed in the stripping process. The organic material can be effectively removed using plasma processes. Depending on the wavelength used for lithography, various additives and solvents are incorporated into the photoresist formulation; for instance, PACs such as diazo-based materials and photoacid generators such as onium salts are typically used. Fortunately, most of these additives are easily stripped in both oxidizing and reducing chemistries.

The stripping and cleaning processes require high-purity gases; typically, 99.9999% purity is used. In-line purifiers have been employed to further reduce unwanted background contamination, although oxidizing plasma stripping processes have a high tolerance for contaminants such as H_2O and H_2 . Reducing processes are also forgiving. However, for Si wafer conditioning, reducing chemistries that contain H_2O and O_2 are easily plasma activated and can oxidize sample surfaces. Most IC manufacturers rely on bulk delivery of gases with appropriate filters and purifiers. Mass flow controllers are used to monitor and deliver precise quantities of gases into the reaction chamber.

Oxidizing Chemistries: Oxidizing chemistries are typically composed of O_2 plus a carrier gas and additives. The oxidizing species can be O_2 , H_2O , or N_2O or other gases containing an O atom. Because photoresist is composed of

hydrocarbons, the O_2 plasmas generate O, which easily reacts with this film to form volatile byproducts:



Small amounts of N_2 (3–5 vol%) have been shown to promote dissociation of O_2 to increase the stripping rate [138,142], whereas adding more N_2 decreases the etching rate. Oxygen is also used to react with implanted species, creating the stable oxides of P: P_4O_{10} , As: As_2O_5 , and B: B_2O_3 . These residues can be rinsed off the surface or cleaned in a subsequent SPM bath.

The stripping rate achieved by using O_2/N_2 chemistry is higher than that attained with the H_2/N_2 chemistry, under similar process conditions. Researchers have shown that the difference becomes more significant at high temperatures [142]. The measured activation energy for photoresist strip using O_2/N_2 plasma is ~ 10 kcal/mol, which is higher than ~ 5 kcal/mol for both 30% N_2/H_2 and the 100% H_2 chemistries.

Reducing Chemistries: Hydrogen (H_2) plasmas are used to remove photoresist and some residues. Although H_2 has not been used to the extent of O_2 for photoresist removal because of the lower stripping rates, the driver for use of H_2 in photoresist stripping and organic contamination removal is elimination of the undesirable oxidation that occurs with O_2 plasmas. The likely mechanism for hydrocarbon removal in H_2 plasmas is similar to that of O_2 plasmas. For instance, H radicals formed in the plasma abstract H from the photoresist and generate free-radical sites, which react further to fragment the photoresist. Atomic H also attacks double and triple bonds in the hydrocarbon materials. When saturated hydrocarbons have formed, they desorb from the surface, consistent with the observation that atomic H has the ability to etch graphite [139]. Thus, it stands to reason that hydrocarbon-based photoresists can easily be removed by hydrogenation to form volatile CH_4 or higher order hydrocarbons. Any plasma reactor designed for photoresist stripping can use a H_2 -based process. However, in most cases, the stripping rates in a H_2 plasma will be lower in a MW system because of reduced ion bombardment. Therefore, additives are used. Pure H_2 can be used to increase the stripping rate. However, forming gas (4–10 wt% H_2 in N_2) is usually a safer alternative than the use of H_2 . Furthermore, the formation of NH radicals in forming gas plasmas, which yields $-CNH$ moieties as reaction products, appears to assist the decomposition and volatilization of photoresist materials [140]. There are limitations when using H_2 as the resist stripping gas because of the plasma source ion energy. High ion energy with this plasma chemistry will cause polysilicon pitting from ion bombardment [141]. By limiting the H_2 concentration to <10 vol%, damage will be minimized. In addition, the hardware configuration is of great importance to enlarge process window. These hardware requirements are outlined in Section 7.4.3.3.

Hydrogen is also used as an additive to oxidizing chemistries. The addition of H_2O to an O_2 plasma can reduce oxidation of Si or metal surfaces. However, if complete elimination of oxide growth is required, only an H_2 -based plasma is appropriate. Furthermore, the addition of a minority component of H_2O vapor increases the concentration of H and OH radicals, which can enhance stripping rates [142].

Researchers have characterized the effects of various downstream plasma chemistries including 100% H_2 , H_2/N_2 , and O_2/N_2 [143]. They have found that the addition of N_2 to H_2 plasma maximized the photoresist stripping rate when the mixture is composed of $\sim 30\%$ – 40% N_2 , about three times that of pure H_2 . For H_2 -based chemistry, a small N_2 addition was found to not only increase the stripping rate but also significantly suppress the Si etching rate.

Additives Chemistry: Additives for stripping and cleaning can be F-based. Inert gases are also added in some cases. Although H_2O is considered an additive, as is H_2 in an O_2 plasma, or O_2 in a H_2 plasma, these are actually components of the reaction process. Additives, such as CF_4 , have been shown to modify the main constituent of the plasma (O) to increase the etch rate.

Fluorine additives are varied: NF_3 , SF_6 , CF_4 , and other fluorocarbons can be used as a source of F in both O_2 and H_2 -based stripping formulations. In some cases, these additives can be used not only to enhance the strip rate but also to remove unwanted residue during the stripping process, thereby cleaning the surface. The drawback of using F-based additives is that F etches Si, SiO_2 , and some metals, thus removing film and substrate layers that may not be desirable or tolerable. Fig. 7.3-23 shows the selectivity of photoresist to SiO_2 with various amounts of CF_4 added to the stripping chemistry formulation [76]. Note that the stripping rate also changes with the addition of these additives.

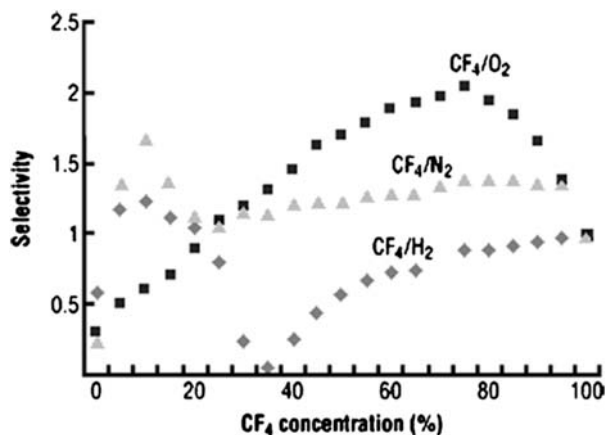


FIGURE 7.3-23 Selectivity of photoresist to SiO_2 with respect to various plasma stripping chemistry formulations [76]. Used with permission from Solid State Technology.

The addition of F-containing gases into the stripping formulation promotes surface cleaning, as already discussed for removal of SiO_x -based residue, implant residue, and metallic residue that is volatilized after complexation with F. Usually, a H_2O rinse is required after cleaning.

Additives may also be inert. Helium, Ar, and N_2 are the most common. If increased ion bombardment is needed, Ar is a good choice; if increased cooling is needed, He is a good choice, but high-energy photon emission from these plasmas may also affect underlying layers. However, these gases are electropositive and may affect the plasma behavior.

7.3.3.2 *Chemistries for Cleaning and Surface Conditioning*

Sputter cleaning and removal of surface oxides are typically performed before thin film deposition. Both Ar sputter-etching and cleaning and H_2 -based oxide removal will be discussed in this section.

Oxide removal from Si or metal surfaces occurs much more slowly than does C removal from the same surface. Because the Si—O bond strength and the O—H bond strength are comparable (~ 111 kcal/mol), the reduction of Si—O is not energetically favored and high kinetic barriers may exist to inhibit H reduction of the oxide. However, plasma processing increases the energy of the reactant species, thereby allowing H-termination of the Si surface. Also, ion bombardment breaks bonds and thus creates Si dangling bonds at the surface where H radicals can react. Once formed, the H-terminated Si surface is in general much less reactive than the SiO -terminated surface. Therefore, an H_2 plasma treatment provides surface passivation from ambient O_2 and also protects the surface before processing the Si wafer at other steps [112]. Ishii et al. [144] have shown that a H_2 plasma treatment generates minimal surface roughness and the H formed penetrates into the bulk Si.

Copper films that have been oxidized by wet processing steps and passivated with organic complexing agents can be cleaned through the use of H-based plasma processes, and then passivated with H_2 , although etching of Cu in H_2 plasmas at low temperatures can occur readily [145,146]. The Cu surface can subsequently be regenerated by heat treatments. This unoxidized Cu surface ensures low contact resistance to a subsequently deposited Cu or other metal layer.

Argon sputter etching has been used to clean residues and unwanted oxide films from surfaces. The relatively large atomic mass of the Ar atom results in significant momentum transfer and thus sputters away the unwanted film. Impurities such as O_2 decrease the sputtering yield [147].

7.3.3.3 *Temperature*

Photoresist stripping at low temperature has a number of advantages. For instance, ashing residues are kept in a H_2O -soluble form, such as chlorides or

fluorides, which in the case of high temperatures could revert to insoluble compounds. Such ashing also helps to minimize the thermal budget during device processing, and thereby aids in the retention of critical electrical performance. Moreover, lower temperature helps avoid the popping that can occur from stripping high-dose implanted resists at temperatures above the hard bake temperature. Sodium and other mobile ions do not diffuse readily into the SiO_2 layer at lower temperatures, which prevents potential gate oxide integrity and field oxide stability problems. Savas has reported that overstripping after the photoresist is removed and the wafer is processed for additional time increases the Na concentration [26]. Fujimura also reported that heavy metals in the photoresist could be driven into the Si substrate at temperatures greater than 300°C [148].

Although temperature effects can be beneficial to enhance photoresist removal rate, it is important to consider all films that will be exposed to the plasma and determine if certain temperature regimes alter the etch selectivity. For instance, the rate of H_2 -based stripping of 193 nm resist materials increases substantially above 200°C; however, the rate decreases if too high pressure is used [149] and the rate decreases if too high pressure (>800 mTorr) is used in this ICP downstream reactor. The decreased removal rate was attributed to hydrogen atom recombination. Hydrogen atoms also etched silicon-based materials, but temperature effects were not as important. The silicon etch rate in the ICP reactor was low at temperatures >40°C, again, apparently because of atom recombination. Silicon nitride etch rates were negligible below 100°C, and even above 250°C, the rate was <0.1 nm/min.

7.3.3.4 Pressure

Although plasmas can be generated at atmospheric pressure, typical plasmas used in wafer processing are performed at subatmospheric pressures. The type of plasma sources, the material to be processed, and the application assist in determining the pressure to be used. The higher the pressure, the higher the gas flow needed to maintain the pressure and the less directionality achieved for the same power and bias conditions. Of course, lower pressures reduce the gas flow rates needed and enhance directionality. Directionality is important with respect to retaining the CDs of device features when using a reactive gas and to ensure that ions reach the bottom of high-aspect ratio trenches to assist residue removal. RF bias at low pressures, less than 100 mTorr, promotes anisotropy while downstream MW plasmas at pressures greater than 1 Torr yield isotropic nondirectional plasma processes. When RF and MW are used simultaneously, the pressure is typically that used for an MW process, while the RF bias adds some directionality.

7.4 PLASMA STRIPPING, CLEANING, AND SURFACE CONDITIONING EQUIPMENT

There are several types of commercially available plasma process systems used for resist stripping and cleaning. The most common reactor types include inductively coupled sources and both remote (downstream) and biased (parallel plate) RF and MW discharges. Another less common source for resist stripping applications is ECR [53], typically used for in situ H_2 cleaning before metal deposition. Except for downstream discharges, the development of many of these sources for cleaning has been paralleled by their development for dry etching processes. This chapter will focus mainly on a review of downstream RF, MW, and ICP sources and will review the addition of RF bias to augment the downstream component of the reactor. Finally, ECR reactors are described briefly.

7.4.1 Overview of Plasma Stripping, Cleaning, and Surface Conditioning Equipment

The evolution of plasma reactors has a long history. The first reactors used in IC manufacturing, the Reinberg reactor [150,151], were used to etch polysilicon for gate definition. Later, plasmas were also used to remove photoresist [152]. The simplest and historically the first plasma reactor configuration to be applied to IC process technology for etching or photoresist stripping was the barrel reactor [153,154]. This system uses a cylindrical chamber typically made of quartz with RF power applied to external electrodes or a coil to couple RF energy into the low-pressure gas.

Multiple wafers (batch process) are placed in a holder that is generally isolated from the plasma by a perforated (often Al) etch tunnel which serves as a Faraday cage to shield the substrates from ion and electron bombardment. The tunnel improves temperature and thus etching rate uniformity across the batch of wafers and reduces radiation damage. This configuration results in little, if any, ion bombardment and thus the etch reactions are primarily chemical and yield isotropic etch profiles. Barrel strippers also contributed to plasma damage from both charging damage to the device from plasma nonuniformity and from UV irradiation damage. Even with the use of an etch tunnel, the wafer is being irradiated with strong UV light through perforations in the tunnel. Furthermore, photoresist removal rates are low because of low plasma density, the lack of ion bombardment, and the inability to effectively heat substrates. Typically, O_2 -based plasmas have been used to strip resist materials, thereby oxidizing the organic resist materials into low-molecular weight, volatile, product species (e.g., CO, CO_2 , and H_2O). Removal rates can be enhanced by increasing the temperature, and by the addition of halogen-containing vapors or H_2O vapor, both of which promote the degradation of

polymer molecules. However, chemistry-enhanced stripping rate improvement is limited by hardware configuration.

As device sizes decreased and aspect ratios >3 became the norm, anisotropic etch profiles were required. In the late 1970s, parallel plate reactors became the workhorse for plasma-assisted pattern definition [93,155]. Because substrates reside within the plasma, resist stripping and cleaning in such reactor configurations are problematic because of direct radiation (ion, electron, and photon) damage and surface charging. Therefore, in the mid-1980s, downstream, sometimes referred to as afterglow or remote reactors were introduced for resist stripping and isotropic, chemically selective etch processes. In these configurations, a substrate is placed on a temperature-controlled platen located downstream of the plasma, which is generated by RF (13.56 MHz) or MW (2.45 GHz) excitation. The tube connecting the plasma source and the substrate is frequently bent to eliminate line-of-sight transport of plasma species and photons from the plasma to the substrate. This configuration promotes electron and ion wall collisions and so recombination, thereby minimizing or eliminating ion, electron, and photon impingement on substrate surfaces; thus, only long-lived neutral species reach the film/substrate surface. As a result, radiation damage to devices is greatly reduced with downstream strippers.

Batch (multiple wafer) processes were used initially in plasma processing to maximize production throughput. The continuing increase in wafer size along with the improved process control needed to achieve within-wafer and wafer-to-wafer uniformity at small feature sizes resulted in single-wafer approaches to etching and ultimately to resist and residue removal. Throughput is a concern in single-wafer systems; therefore, high strip rates are required. This is normally achieved in downstream configurations by establishing a high neutral species concentration with a uniform flux to the substrate surface while maintaining a short residence time. Spectroscopic studies of afterglow or downstream plasmas have been reported [156,157]. Further discussion of the diagnostics of these plasmas is discussed in Section 7.5.

HDP systems such as ECR, ICP, helical resonator, and helicon reactors have been prevalent in etching processes for many years. The combined advantage of low pressure, high ionization efficiency, low plasma potential, and separation of plasma generation from ion energy control is most attractive. However, because these sources involve high ion currents to the wafer surface, they have not been used extensively in photoresist stripping applications; their primary utility is in surface cleaning and preparation and, of course, patterning.

7.4.2 Remote Plasmas: Downstream Generation

Typically, in a remote plasma process, a reactant gas is introduced through the plasma tube where it is ionized and/or dissociated by the applied plasma

power. Operating pressures are usually in the range of 100–1000 mTorr. Reactive species such as atomic O generated by electron impact dissociation of O₂, along with ions, radicals, and electrons, are generated in the plasma-coil region, generally known as the “glow discharge” region. Convection and diffusion processes transport these species from the plasma region to the substrate surface thereby leading to the term “downstream.” In some instances, an afterglow extending from the plasma coil toward the substrate is observed under typical pressure and power conditions, although this is not always a desirable situation.

The plasma tube, constructed of quartz, silica-alumina ceramic, or sapphire, is an important component, because the optimum material choice depends on the gas mixtures used and the plasma frequency. Fluorine-containing gas mixtures will etch quartz tubes. Sapphire and silica-alumina ceramic are inert to F-containing plasma chemistries, but are much more expensive than quartz. The diameter is also important, because for larger tube diameters, higher volumetric gas flow rates can be used. Thus, smaller diameter tubes have flow rate and pressure restrictions that may limit the plasma-operating regimes.

Elbow bends in the downstream reactor system are also important because one 90-degree bend is usually sufficient to eliminate charged species and unwanted photons that may cause electrical damage to the wafer. In addition, materials of construction for the gas distribution showerhead and the geometrical configuration of the holes in the showerhead are important design considerations.

7.4.3 Commercially Available Plasma Equipment

Remote or downstream plasma processing has been implemented for a variety of semiconductor single-wafer processing steps and is reasonably well understood and characterized. Parallel plate reactors for cleaning applications were available since the late 1970s, but did not gain widespread acceptance in the late 1990s. Barrel-type batch reactors were the first commercially successful plasma stripping systems, gaining popularity because of the high throughput and low damage. These systems were used throughout the 1980s, in the 1990s single-wafer downstream reactors replaced batch reactors because of improved uniformity and the ability to control radiation damage by incorporating a 90-degree bend in the plasma tube as opposed to the use of a Faraday cage to prevent ion bombardment damage [158]. Today, plasma stripping systems have high throughput, low damage, high uptime, and low cost of ownership, especially when compared with other wafer-processing equipments such as plasma etch, CMP, and plasma deposition systems.

The frequency of the power for plasma generation was not clearly defined when remote plasma photoresist stripping was first introduced [158,159]. The frequencies are now based on Federal Communication Commission acceptable

ranges that do not interfere with communications and are outlined by standards set by the Industry, Scientific, and Medical RF equipment guidelines, CNS13803. Studies have been undertaken to determine the optimum frequency for plasma efficiency, as discussed in [Section 7.5.1](#). Initially, RF sources were used for remote plasma generation; however, in the 1990s MW was the frequency of choice for downstream remote plasma reactors while ICP downstream reactors typically use RF sources. For advanced logic and memory application after 2000, ICP is used as the plasma source in many commercially available tools because of its capability to enable higher stripping rates compared with other plasma sources.

7.4.3.1 Downstream Radio Frequency Plasma Sources

RF plasma sources were used initially for barrel-type photoresist stripping reactors. With the evolution to single-wafer technology, RF continues to be used, although with the large cavities needed for plasma generation and thus the higher power and lower pressure for generation, MW frequencies became attractive because of the normally higher photoresist strip rates. The plasma is generated through capacitively coupled plates on either side of the plasma tube or chamber. The plasma then flows downstream through a gas distribution showerhead to the wafer. The long-lived plasma species etch the photoresist and the byproducts are exhausted through the vacuum port. RF downstream plasma sources have been replaced by MW sources because of the generally higher efficiency of plasma generation, in the 1990s time frame.

Sputtering of the chamber material on the wafer is a concern anytime high-energy ions are used for etching. Aluminum contamination is of particular concern because the majority of the process chambers are composed of aluminum or alumina. Where possible, plasma systems used for IC manufacturing should avoid stainless steel components (chamber or fastener material) because the components of stainless steel (Fe, Ni, and Cr) cause electrical failure of devices. In extreme cases, quartz liner must be used to satisfy the most stringent contamination requirements for <10-nm devices.

7.4.3.2 Downstream Microwave Plasma Sources

The basic design that is typical of a MW remote plasma stripping and cleaning system is shown schematically in [Fig. 7.4-24](#). MW sources normally operate with resonant cavities, which provide higher plasma densities than RF plasmas operating in a capacitively coupled mode. Downstream plasma processes and configurations are inherently isotropic in nature, allow charged particles to be removed from the gas flow, but permit neutral species to flow unimpeded to the wafer surface.

In most downstream MW plasma reactors, the plasma is generated in a tube that is inserted into a MW cavity. Gases within the tube are excited by MW power coupled to the gas, thus forming the plasma. The plasma flows

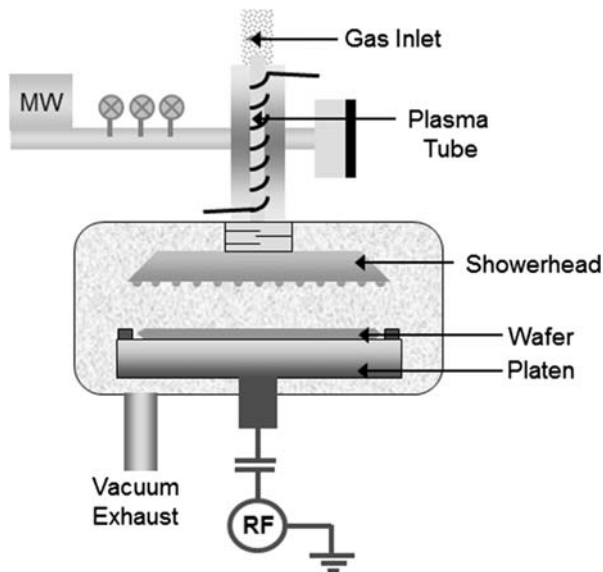


FIGURE 7.4-24 Basic design of an microwave remote and radio frequency bias plasma stripping and cleaning system with the key components designated.

downstream to the wafer residing in the process chamber. Plasma tubes are typically composed of quartz (SiO_2), sapphire (Al_2O_3), or composite ceramics ($\text{SiO}_x/\text{AlO}_y$). When F-based chemistries are used, sapphire or ceramic tubes are required, as quartz tubes show significant erosion. Quartz plasma tubes usually show higher removal rates than do sapphire or ceramic tubes because of the reaction of atomic O and F with these materials. Sapphire tubes are more prone to thermal stress than quartz tubes because of their higher thermal expansion coefficient, limiting the MW power that may be applied to the tube. However, sapphire tubes can be cooled to prolong their lifetime. One-piece sapphire tubes are preferable because of the manufacturing issues caused by creating the tubes. In some cases, a 90-degree bend is included in the plasma tube to eliminate the need for baffles to prevent direct ion bombardment and line-of-sight UV irradiation. Bends in sapphire tubes are much harder to manufacture than are bends in quartz tubes.

For processes requiring directionality, which downstream plasmas do not provide, a separate RF-powered biased electrode is used to accelerate ions toward a wafer surface. Placing the wafer on a heated pedestal or using infrared (IR) radiation regulates the temperature. Raising the wafer on pins above the heated platform is one method of wafer cooling for lower temperature processes; however, once raised, the RF bias is not effective.

Typical chamber materials are made of Al. Because the stripping and cleaning plasmas usually do not reach the wall of the reactor, sputtering is minimal. Because atomic O is very chemically reactive, it recombines rapidly

on the surface of metals used for chambers in cleaning and stripping. The nature of these reactions and the factors that influence the effects are not well understood, so manufacturers have been discouraged from using Al chambers for MW stripping with atomic O because of contamination concerns. However, quartz-lined chambers have been successfully used to pass the most stringent contamination requirement and is the industry standard for devices that require low metal contamination levels.

There are two major issues for MW downstream strippers in the production environment. The main issue is high cost of the MW generator quartz tube consumable. Because of the erosion caused by the plasma generation, the tube must be replaced frequently. The secondary issue is that the resist stripping rate in an MW downstream system is still not as high as in an ICP. This is because of the MW downstream plasma-generated species having a very long path from the plasma source to the substrate. Along this path, many of the active species generated at the plasma tube recombine into nonreactive species by the time they reach the wafer surface; thus, the stripping rate is greatly reduced. Although still in use, MW downstream plasma systems have been replaced by ICP systems for <45-nm devices.

7.4.3.3 Inductively Coupled Plasma Sources

The basic design of an inductively coupled remote plasma cleaning system is shown schematically in Fig. 7.4-25 for a plasma direct contact configuration. The system has a spiral induction coil wound around a quartz tube where the plasma is generated; RF power applied to the induction coil generates a plasma. A Faraday shield can be placed between the induction coil and the

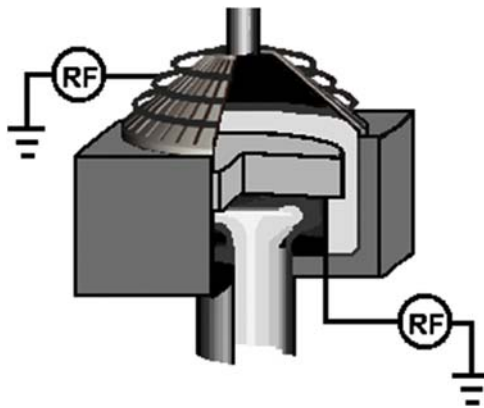


FIGURE 7.4-25 Basic design of an ICP remote and radio frequency bias plasma stripping and cleaning system with the key components designated. Adapted from S. Savas, R. George, D. Gilbert, J. Cain, M. Herrick, A. Nagy, K. Karuppana, *Micro*, 22 (8) (October/November 2004) 61. Used with permission from Mattson Technology.

reactor chamber to block capacitive coupling of energy into the reactor chamber, thus mitigating secondary plasma formation from the direct coil electric field coupling and minimizing the ion energy of the charged species that flow downstream to the wafer surface. For processes that require enhanced ion bombardment, a bias RF-powered electrode may be included allowing the wafer to be directly exposed to the plasma.

There are two different types of ICP systems. The first type is where the plasma has direct contact with the wafer to form a plasma sheath, plus an optional RF bias to control ion energy. In this case there is no major difference between an ICP plasma etching chamber and the stripping chamber. In this configuration, the wafer remains at a temperature $<100^{\circ}\text{C}$. Because of direct plasma-induced heating from the plasma contact, the wafer can rapidly heat up if there is not an electrostatic chuck to maintain a low temperature. The second type of ICP system uses a “grid” or “baffle” to separate the plasma generation chamber to the wafer reaction chamber. The usage of the grid or baffle reduces the charged species while allowing the active neutral species to reach wafer surface. Both types of IPC systems provide significant stripping rate improvements over MW downstream systems.

The grid can be made from insulating materials, such as quartz or ceramic, or from conducting materials, such as Al. When only a single grid is used, both ionized species and UV are greatly reduced, but nevertheless, some species reach the surface. Therefore, multiple grids with offset holes can be used to eliminate the majority of the ionized species while further reducing the UV exposure. [Fig. 7.4-26](#) shows the ion probe measurement directly above the wafer surface. The ion current reaches close to a background value, for Grid 2. In this example, the wafer is in total darkness because the metal grid blocks all the UV originating from the plasma generation chamber.

The Faraday shield placed between the RF coil and the quartz tube used for plasma generation is another key feature for advanced ICP systems. There are several purposes for this configuration of the Faraday shield. The first purpose is to reduce the electrical field coupling to the plasma to reduce the local erosion as shown in [Fig. 7.4-27](#). The second purpose of the Faraday shield is to reduce the ion energy in the plasma generation so a softer etch can be achieved to reduce possible substrate damage. [Fig. 7.4-28](#) shows the ion energy distribution with and without a Faraday shield. This feature is especially important for H_2 plasma applications. If the ion energy is too high, it will limit the H_2 vol% that can be used. In particular, pitting generated from higher energy bombardment from hydrogen plasma is an issue for application such as polysilicon gate cleaning.

ICP plasma systems take advantage of the inductive coil to control the ion density and plasma potential while the powered electrode may be used to independently control ion energies for residue removal. Optimization of the chamber and coil configuration produces a highly uniform plasma potential across the surface of the substrate leading to uniform ion bombardment. In

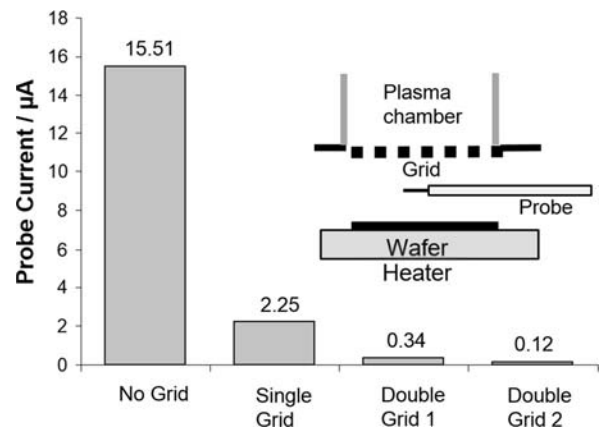


FIGURE 7.4-26 Schematic of a plasma current probe inserted into an IPC chamber after the chamber grid. The graph shows that the intensity of the current, and thus the ionized species, is reduced by use of a grid. Adapted from V. Vaniapura, S. Ma, B. Elliston, H.-A. Phan-Vu, D. Liou, K. Kang, *A Novel Post High Dose Implant Strip (HDIS) Process with Controlled Crust Removal for 28nm and Below Logic Technology*, Semicon China 2017 CSTIC (Shanghai, China), 2017, pp. 16. Used with permission from Mattson Technology.

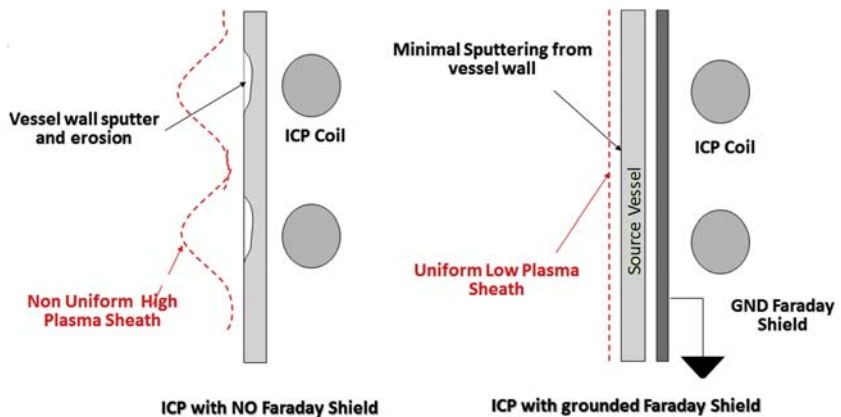


FIGURE 7.4-27 Schematic shows the effect of a grounded Faraday shield used to reduce the erosion of the vessel wall. The erosion is caused by electric field coupling to the vessel from the ICP coil. Adapted from V. Vaniapura, S. Ma, B. Elliston, H.-A. Phan-Vu, D. Liou, K. Kang, *A Novel Post High Dose Implant Strip (HDIS) Process with Controlled Crust Removal for 28nm and Below Logic Technology*, Semicon China 2017 CSTIC (Shanghai, China), 2017, pp. 16. Used with permission from Mattson Technology.

addition, one ICP supplier uses the conical chamber and coil configuration to produce reactive neutral species at varying distances within the chamber volume for nonion enhanced processing. This configuration provides uniform diffusion of the activated neutral species across the wafer surface and thus

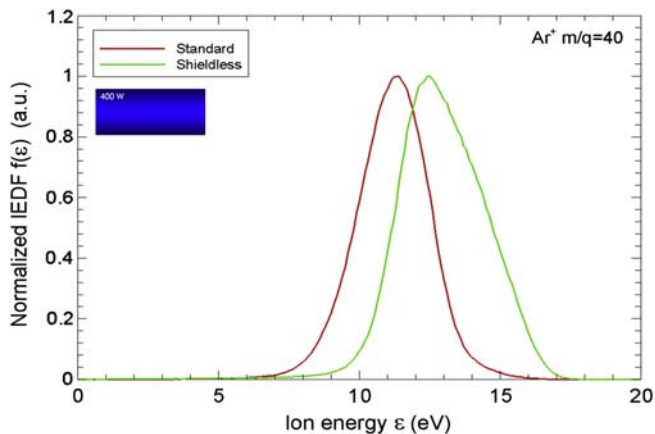


FIGURE 7.4-28 Effect of a grounded Faraday shield (standard) compared with no Faraday shield (shieldless) with respect to ion energy distribution. The Faraday shield shows lower ion energy results from its use. Adapted from V. Vaniapura, S. Ma, B. Elliston, H.-A. Phan-Vu, D. Liou, K. Kang, *A Novel Post High Dose Implant Strip (HDIS) Process with Controlled Crust Removal for 28nm and Below Logic Technology*, Semicon China 2017 CSTIC (Shanghai, China), 2017, pp. 16. Used with permission from Mattson Technology.

uniform stripping and cleaning [160]. Fig. 7.4-29 shows the ion density of the ICP reactor with respect to the applied power to the coils [75]. A similar design to the ICP is the I²CP, interlaced ICP. In this case the coils intertwine and there is less need for a showerhead or a Faraday shield. An example is shown in Fig. 7.4-29.

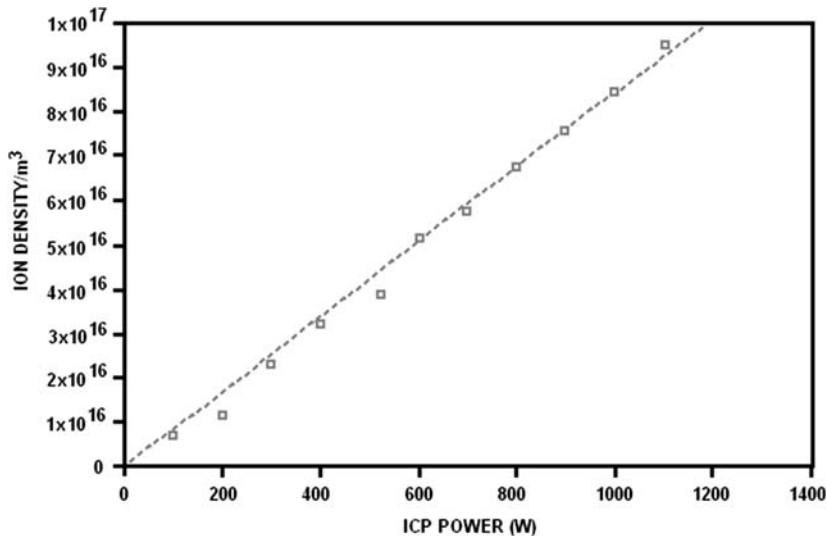


FIGURE 7.4-29 Ion density as a function of applied ICP power [75]. Reprinted from *Micro Magazine*. Copyright 2004, Tom Cheyney, Los Angeles, CA. Used with permission.

7.4.3.4 Parallel Plate Plasma Sources

The basic design of a typical parallel plate plasma cleaning system coupled with a downstream MW source is shown schematically in Fig. 7.4-30. For cleaning applications the parallel plate is combined with the downstream source to create a biased flux of charged particles that reaches the sample in an anisotropic (directional) manner. RF-biased plasmas alone are rarely used for photoresist stripping; however, the combination of downstream plasma generation and RF bias, also known as RIE gives faster etch rates when the wafer is directly exposed to plasma. The use of bias also assists in the removal of residue by ion bombardment, without which certain residues cannot be volatilized [100].

RIE addition also gives the ability to perform etching in the stripping and cleaning system. Etching of dielectric barrier films such as SiC or SiN_x after the low-k dielectric post-etch clean eliminates the need to both remove the wafers from the stripping and cleaning system and transfer the wafers to an etch system for final barrier etch. Typically, only “soft” etches are done in the plasma stripping system; however, biasing the wafer at higher powers allows higher etch rates and thus expands the capability of formerly resist stripping- and cleaning-only tools.

Typically, the wafer sits on a temperature-controlled platen. Water or H₂O-based cooling liquid is circulated to control the wafer temperature. Sometimes

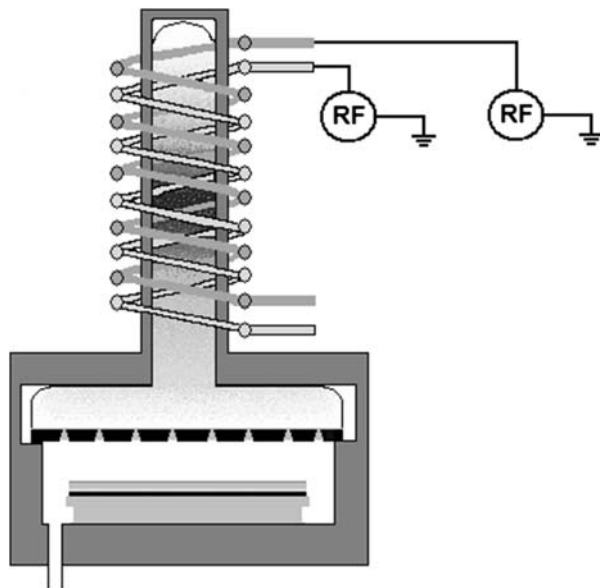


FIGURE 7.4-30 The basic design of an I²CP remote stripping and cleaning system with the key components designated. *Used with permission from Novellus Systems.*

an electrostatic chuck is used to cool the wafers for resist stripping and cleaning at temperatures $<60^{\circ}\text{C}$.

7.4.3.5 Electron Cyclotron Resonance Plasma Sources

An ECR source can be used to generate high-HDPs. ECR sources use MW excitation (2.45 GHz) in a resonant cavity similar to that of ICP; however, the plasma is formed directly over the wafer, not upstream from the wafer. Activation efficiency is enhanced by magnetically confining the electrons in the plasma chamber. With ECR, a discharge can be sustained at a pressure of approximately 2 orders of magnitude lower (2×10^{-4} Torr) than that used for remote RF chamber volumes. Neutrals and charged particles flow out of the plasma chamber toward the sample stage. Interactions between charged particles and the substrate can be controlled through RF bias voltages applied to the substrate. ECR sources permit H-based cleaning with high atomic H flux and higher residual gas pressure at lower cleaning temperatures [161]. Researchers have demonstrated Si surface cleaning and surface conditioning by passivating the surface with H before epitaxial deposition [162,163].

Hollow cathode plasma arrays have been proposed for photoresist stripping using H_2/N_2 mixtures [164]. This configuration generates a high-density uniform discharge in a 7:3 mixture of H_2/N_2 at 1.6 Torr, with a photoresist removal rate of 260 nm/min at 250°C .

7.4.3.6 Other Plasma Sources

Although it is beyond the scope of this book, many other plasma sources exist. For instance, arc reactors, hot-wire technology [166], atmospheric and near-atmosphere pressure plasma torches [165–168], atmospheric and near-atmospheric pressure plasmas [156] are being developed for various cleaning and surface conditioning processes. Researchers are investigating these technologies and if they prove to be cost effective with improved processing results, they may ultimately be implemented into production. Specifically, photoresist stripping rates greater than $10\text{ }\mu\text{m/min}$ at low temperatures, $<200^{\circ}\text{C}$, have been demonstrated with atmospheric O_2 -based plasma processes [169]. However, uniformity of plasma flux and the high flow rates of inerts (e.g., He and Ar) required are concerns with atmospheric pressure reactors.

7.4.4 Auxiliary Equipment and Components

In addition to the source assemblies required for generating the plasma, many of the plasma module design constraints are generic to all remote plasma configurations. Important assemblies include ultrapure gas delivery systems, vacuum systems, wafer transfer, substrate heating, and endpoint detection. Although it is beyond the scope of this chapter to discuss all the components and assemblies required for commercially viable plasma systems, those that are key to the system will be briefly reviewed.

Ultrapur gas delivery is important to assure that contamination is not introduced into the reaction chamber. Stainless steel electropolished tubing is used because most gases do not react with this material. Corrosive gases, when used, typically employ double-walled, protective tubing. In addition, most gases are filtered before use. Gases are usually remote from the plasma system, although point-of-use H_2 generators are available to avoid transporting highly flammable gases from the remote skid to the process system.

The vacuum system typically invokes a vacuum port leading to a remote vacuum pump, sometimes equipped with a roots blower. The vacuum ports are usually arranged radially around the bottom of the chamber to achieve uniform evacuation of gases—both the reaction products and the reagent gases. Typically, the chamber is pumped down to base pressure when the chamber is not being used. Load locks have their own vacuum pumps, and because of the lower volume, the pump speed need not be as high as that for the reaction chamber.

Contamination from wafer movement with the use of end effectors is common in all types of processing equipments. In particular, contamination of the back surface of wafers by robotic transfer usually occurs at the contact points of the end effectors. The contact points have been minimized as robots and transfer mechanisms become more sophisticated. End effects that use edge grip are becoming more common. Cleaning of the end effectors is a routine preventive maintenance procedure and is scheduled regularly to remove unwanted material that has inadvertently found its way onto the end effector. Movement of the wafer is designed to avoid scraping against surfaces; gentle movement of the wafer when placed on and removed from the platen eliminates unwanted contact that could generate particles. Movement in and out of the front-opening unified pod also is optimized so as not to generate particles from the movement or abrasion at the touch points.

Heating of the wafers has been accomplished with several types of in situ radiant or thermal heaters. Quartz-halogen lamps have been used to irradiate the back or front surface of Si wafers either directly or indirectly. Wafers are indirectly heated by first heating a platen in physical contact with the wafer. Alternatively, resistive heaters are also used in situ, as are water or other fluids that circulate within the wafer platen. Although not common, electrostatic chucks that control the wafer temperature are used in some commercially available systems.

Load locks are used on some commercial plasma stripping systems, targeting wafer cleaning, to reduce atmospheric contamination, especially moisture, from entering the reaction chamber. It has been demonstrated for advanced technology development and manufacturing, load locks are a necessity for contamination reduction, especially in the presence of H_2 chemistry. However, most of the plasma stripper systems in the packaging industry do not use a load lock, primarily to reduce the tool cost. Depending on the system configuration, applications, and the gases used, the load lock can also prevent unwanted toxic gases from escaping the reaction chamber.

7.4.5 Safety

Care is taken to prevent accidental release of H_2 into the atmosphere by the use of load locks, double-enclosed gas lines, and gas line interlocks. The percentage of H_2 gas in reducing gas mixtures usually does not exceed 4 vol% to ensure safe operating conditions. However, higher amounts of H_2 are being used in reaction chambers, as high as 100 vol% has been shown in some of the process recipes in advanced plasma stripping application [142]. Hydrogen gas is delivered either through the use of gas cylinders within enclosed cabinets or by point-of-use H_2 generators.

Fluorine-containing compounds used to increase the etch rate of photoresist and to remove post-etch residue are ambiguous with respect to safety, CF_4 is inert and requires high temperatures to combust, NF_3 is highly toxic and a highly reactive molecule, while SF_6 is inert but reacts at lower temperatures. However, because all F-containing vapors are dissociated in the plasma, F atoms are generated which can react with H-containing species in the plasma to form HF. Care must be taken with F-containing gases to avoid releases to the atmosphere and to ensure that the vacuum effluent is exhausted and effectively scrubbed.

Another safety concern is erosion of the quartz tube when using F-containing gases. The tube may thin to the point it becomes mechanically unstable. With <1% vol of F used, in general, is compatible with quartz tube. However, if higher F concentrations are used, ceramic tubes are recommended for better erosion resistance, but with the plasma coupling also undesirably being reduced.

7.5 PLASMA DIAGNOSTICS

7.5.1 Endpoint Detection and Plasma Diagnostics for Stripping and Cleaning

Detection of the endpoint, the point at which either the photoresist or the residue has been completely removed from the wafer is used routinely in IC production. Removal of the photoresist and residue without overprocessing minimizes plasma damage and optimizes wafer throughput. Reproducibility of the endpoint is critical for control of high-volume production. Endpoint detection will be discussed in this section. In many cases, monitoring endpoint during production environment is more important than using endpoint to control the overash percentage because the endpoint time can be used as production monitoring to detect potential yield issues or line yield problems.

Plasma atmospheres supply chemically reactive species and ion bombardment for photoresist, residue, and contaminant removal. However, the complexity of the chemical and physical processes that occur during such processes is considerable. As a result, use of in situ plasma diagnostics to establish fundamental understanding and thus prediction of specific reactions and removal rates is difficult to achieve; it is unlikely that this level of understanding will be generated any time soon. At the very least, careful control of process parameters (e.g., power, pressure, flow rates, and gas composition) is required, but this is often insufficient for adequate process control. For instance, innumerable chemical reactions take place in the gas phase and at surfaces during plasma processing. Plasma and film uniformity across wafers and from wafer-to-wafer display some variability: reactor wall conditions determine the rates of destruction of etchant species; outgassing, residual leaks, and small concentrations of impurities in feed gases can alter the chemistry that occurs in reactor systems. Implementation of production-worthy photoresist stripping and cleaning processes has therefore depended upon the development of appropriate diagnostic tools and endpoint detection methods.

Highly sophisticated diagnostic techniques are available to investigate chemical species concentrations, ion fluxes, and ion energies, but such methods are typically incompatible with production systems. Instead, endpoint detection methods, which do not require nor necessarily generate fundamental understanding of the process, can serve as monitors of film/contaminant removal. That is, a change in plasma parameters, in relative reactant or product species concentrations, or in film composition at the endpoint is generally adequate to permit reproducibility, control, and optimization of processes. A large number of approaches to diagnose plasmas or provide endpoint detection have been explored over the past 30 years such as optical emission spectroscopy (OES), laser interferometry, and measurement of control parameters such as temperature, pressure, or RF power. Detailed description of the various methods is beyond the scope of this chapter. However, the principles behind the primary methods used (OES, mass spectrometry, and Langmuir probes) will be presented briefly in [Section 7.5.2](#) where plasma spectroscopy is discussed. Additional information on these and other techniques can be obtained from the extensive literature in this field [\[153,170–172\]](#).

Stefani et al. shows an example of implementation of process monitoring of photoresist clearing times on a wafer-to-wafer basis using OES on a plasma stripping system [\[173\]](#). Control of the process is made more difficult because prior processing steps such as ion implantation affect the properties of the photoresist material and consequently the stripping time.

7.5.2 Optical Emission Spectroscopy

Light emitted by the plasma or from the reaction of the photoresist, residue, and the plasma gases include visible, UV, and IR light. This light can be used to monitor the reactions involved in film removal because of changes in the chemical composition of the plasma as the photoresist and residue are removed from the wafer [174]. The plasma is composed of the gas used to etch the photoresist and the byproducts of the plasma reaction, plus any atmospheric gases that may inadvertently reach the chamber, such as N_2 . Many of the gas molecules in the chamber are not in their electronic ground states; rather, they have been excited by the plasma and emit radiation in the form of light, which can be analyzed. The change in the intensity of the light can provide information about the chemical species, their relative concentration, and thus the processes taking place within the chamber. Typically, emission at one or more wavelengths is monitored; each wavelength corresponds to emission from a particular excited state species, thereby allowing reaction progress to be followed. The background and theory of this technique is discussed by Payling et al. [175].

Oxygen MW and RF plasma emissions have been studied [176–179] to assist in determining the optimum process parameters for photoresist stripping. The RF plasma generates more charged species, both O^+ and e^- , than the MW plasma, which generates relatively more atomic O [182]. MW plasmas have been shown to have higher etching rates and less charge damage than RF plasmas. Such results are believed to be a result of the preponderance of long-lived radicals that are able to etch the photoresist more effectively than the shorter lived ions, which also cause charge damage, as discussed in Section 7.6. Collert et al. [157] found that atomic O does not play a significant role in the stripping of photoresist when using an RF discharge at low pressures (75 mTorr) and low flows (10 sccm) of O_2 .

For both MW and RF downstream systems, ionized species directly contact the wafer surface during the process. Therefore, the ionized species of the plasma provides a background of constant light emission. However, for ICP plasma systems that use a double grid with offset holes, most of the ionized species will not reach the wafer surface so the wafer is exposed to extremely low or no background light emissions. In this case, only neutral species reach the wafer surface to react with photoresist. In other cases, once the photoresist is consumed by the active neutral species, the light is extinguished for most of the visual wavelength.

The wavelength of the light used to detect the endpoint is dependent on the reaction chemistry. Some of the reactive gases used in plasma stripping are very dilute and thus make detection of the byproduct or the reaction gas species very difficult. Also, downstream plasmas sometimes have few molecules in excited states, especially near the wafer surface, resulting in detection limitations. The light may be monitored in the reaction chamber or downstream. Build-up of impurities in the reaction chamber may influence the endpoint. If the photoresist resides on an organic-based film, monitoring

TABLE 7.5-8 Examples of Wavelengths Monitored for Photoresist Stripping Taken From Refs. [176,180]

Species Monitored	Wavelength (nm)
CO	297.7
OH	308.9
CO	483.5
CO	519.8
H	656.3

the reaction with OES may be difficult, because the organic film may have the same reaction byproducts as the photoresist. Typically, CO is monitored for photoresist stripping applications using O₂, although other wavelengths can be monitored. Table 7.5-8 shows some of the wavelengths that are generally monitored [176,180].

An example of an optical emission-based endpoint is shown in Fig. 7.5-31. The endpoint is for a high-dose implant strip process using oxygen-based

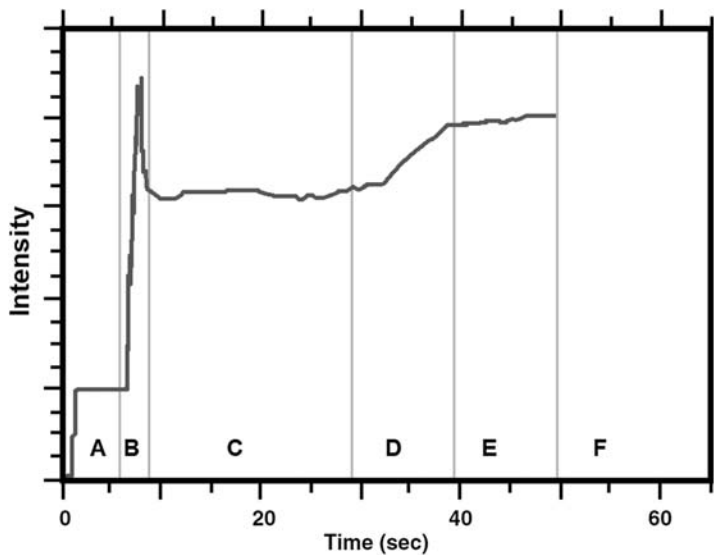


FIGURE 7.5-31 Optical-based endpoint signal [31]. The regions are designated: (A) gas stabilization, (B) plasma initiation, (C) photoresist removal, (D) photoresist removal completion, endpoint is detected, (E) overstrip process, and (F) termination of process. Reprinted from *Micro Magazine*. Copyright 1998, Tom Cheyney, Los Angeles, CA. Used with permission.

chemistry. The various regions are designated, e.g., gas stabilization, plasma initiation, and termination of the process in this case, and can be distinguished by a change in the emission signal. Commercially available process equipment usually comes equipped with endpoint detection apparatus, typically OES. Algorithms are set up to detect the endpoint based on the signal level and rate of change.

7.5.3 Laser and Optical Interferometry

In addition to OES, laser interferometry, also called optical interferometry, is another endpoint technique for plasma stripping and cleaning. Interferometry is based upon detection of the optical characteristics of a broadband or laser source reflected from the wafer surface that provides information on the removal rate of films by measuring the remaining thickness. However, because the light source is directed at the wafer surface, this technique is dependent on the optical properties of the film and the spatial orientation of the features. If the features block the light source, such as high-aspect ratio lines or contact and via holes, inaccurate information is relayed about the removal process. Interferometry may be used to monitor photoresist on opaque materials such as metals, photoresist on organic materials with different optical properties, and also photoresist on polysilicon, SiO_2 , or other films with indices of refraction different than that of the photoresist. As the photoresist is removed, the interference patterns change, either increasing or decreasing in amplitude, depending on the film stack being monitored.

Typically a single light source is used and is directed at a target test site coated with photoresist. An alternative is to use two light sources, one to monitor an area that is free of the photoresist and one to monitor the area with photoresist. An open area of the die on the wafer is used and must be repeatable when monitored wafer to wafer. The laser light must have a direct path to the wafer surface and may require an assembly through the top of the plasma chamber. Interferometry may be used to set up a production photoresist strip recipe to determine the time to endpoint, and then a set-timed process is implemented for the manufacturing process.

7.5.4 Residue Gas Mass Spectroscopy

The gas molecules produced during the resist stripping or cleaning reaction that go into the exhaust can be used to detect an endpoint signal. In this case, some of the exhaust can be redirected into a small enclosed volume. Once enclosed, a small plasma source is used to strike a secondary plasma where the gas molecules of interest are ionized. These species flow into the mass spectrometer for analysis, as shown in [Fig. 7.5-32](#). The mass signal increases as the gas molecule of interest is generated, then decreases until the trigger for the endpoint is signaled. However, a limitation to this method is the delay time.

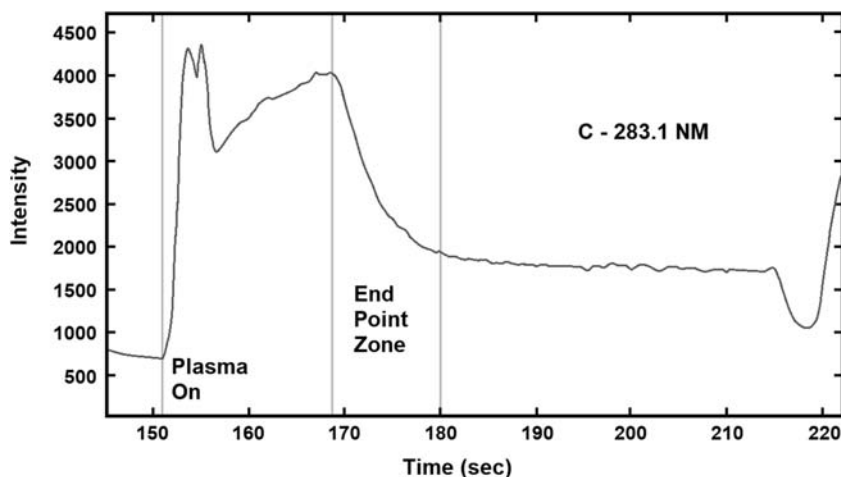


FIGURE 7.5-32 Residual gas mass spectrometer endpoint signal using a secondary downstream plasma to excite the molecule of interest. Adapted from V. Vaniapura, S. Ma, B. Elliston, H.-A. Phan-Vu, D. Liou, K. Kang, *A Novel Post High Dose Implant Strip (HDIS) Process with Controlled Crust Removal for 28nm and Below Logic Technology*, Semicon China 2017 CSTIC (Shanghai, China), 2017, pp. 16. Used with permission from Mattson Technology.

The “real” endpoint occurs on the wafer before the gases flow into the detector; thus, a time delay is experienced. In addition, the unit for secondary plasma generation and mass spectroscopy is more expensive than the regular optical endpoint. Therefore, this analysis technique is less popular for production endpoint, but primarily used to study the plasma stripping reaction mechanism.

7.6 PLASMA DAMAGE

7.6.1 Overview of Damage

Of particular concern with plasma-based stripping and cleaning processes is ion-induced and UV-induced damage [181,182]. Ion-induced damage occurs when ions from the plasma bombard the films causing either electrical damage or physical damage. In charging damage, ions transfer charge to the surface, which if severe, affects the electrical characteristics of the film; this is particularly problematic with thin dielectrics that are subject to breakdown at the high electric fields generated. If the charge approaches the Si substrate, the carrier recombination lifetime can be adversely affected. Of course, the plasma also generates neutral species, but these are not detrimental to the wafer, but are the main contributor to the removal process. In physical damage, ion bombardment causes surface property changes such as film loss, film structural changes, or oxidation. The UV-induced damage occurs because of generation

of light within the plasma discharge; the high-intensity UV radiation can create hole—electron pairs in the semiconductor and can decrease the recombination lifetime by causing charge traps in the semiconductor and/or at dielectric-semiconductor interfaces. Also, UV photons can break bonds in dielectrics and create traps within the film that capture or release electrons or holes and so alter electrical properties. In addition, metallic contamination sputtered from the walls of the reactor or from impurities in photoresist may be driven into the thin films and the substrate because of the high temperature and ion bombardment involved in the photoresist strip process. Fig. 7.6-33 shows the generation of species that may cause damage; damage prevention methods will be discussed later in this section. Other damage mechanisms may be because of electrostatic charging from friction between chamber parts or high ion bombardment energies that can displace bonded atoms; all of these phenomena can affect device yield and reliability.

During the cleaning process, Si or SiO_2 may be etched, which leads to microscopic surface roughening and material loss. Loss of critical material around sensitive features may lead to current leakage. Diffusion of H into the bulk of the crystal is also an issue with H_2 -based plasma processing because of bulk-structural damage and dopant deactivation [183].

As IC manufacturing processes are performed at lower and lower temperatures because of the thin doped junctions and sensitive thin films, plasma-induced damage to substrates will not be annealed during wafer

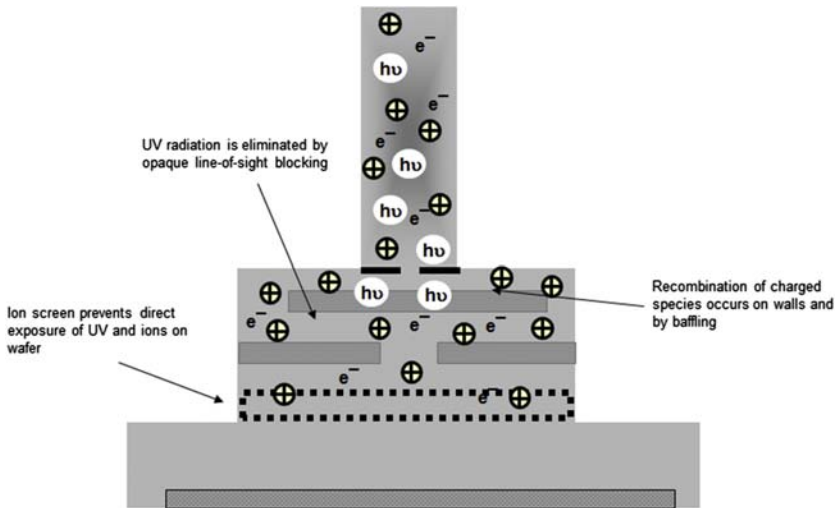


FIGURE 7.6-33 Schematic of possible causes of plasma damage showing the generation of the species that may cause damage, $h\nu$, positively charged ions $\oplus$, and electrons e^- . Also shown are the preventative methods of an ion shield, also known as a Faraday cage, and baffles to prevent direct UV irradiation of the wafer.

processing. Remote plasma stripping and cleaning have not been a major cause of ion-induced damage and charge damage because the electric field strength in the region of the substrate is low, thereby avoiding excessive ion acceleration into the substrate. Although charged species, both ions and electrons, are formed in the plasma, by generating the plasma in a region remote (upstream) from the substrate surface, active chemical and physical species must reach the substrate surface by convection or diffusion. Recombination of ions and electrons and absorption of UV radiation takes place on reactor walls or other surfaces in the plasma system, so that few charged species reach the wafer surface. However, as the device becomes more sensitive because of decreasing film thicknesses and line widths, even a few charged species can enhance undesirable surface reactions to cause either physical or electrical damage.

Traditionally, separation of the plasma generation region from the substrate has been sufficient to minimize high-energy ions that may travel to the substrate. However, to augment this recombination and absorption, Faraday cages and UV-blocking windows have been incorporated into the design of plasma systems. In particular, the implementation of grounded double-metal grids has successfully blocked the UV light and reduced the charged particle species to a low level. The ability to filter these charged particles and thus eliminate completely ion bombardment to minimize surface damage is the key to a production-worthy IC manufacturing process. In general, downstream MW and ICP plasma generation, in which the plasma is formed remotely and the reactive species travel a path with many bends, ensures that ions and electrons collide with the walls before reaching the wafer surface. However, this path reduces the efficiency of the plasma. Therefore, the requirement to generate high-HDP with high power, using an efficient source for the plasma generation to compensate for the decrease in active species, to maintain a high removal rate, has been demonstrated by Savas [184] on an ICP system.

With the incorporation of RF bias, the plasma has direct contact to the substrate; the substrate is surrounded by the plasma sheath similar to plasma damage in an etch chamber. Damage from plasma stripping and cleaning is frequently observed. RF bias can generate charged particles in the plasma above the wafer, resulting in direct bombardment. In a biased-plasma the wafer attracts ions or electrons to the charged electrode, depending on the RF power, frequency, and bias conditions. Typically, a negative potential develops on the wafer surface because of accumulation of negative charge from electron bombardment. The most common root cause of plasma charging is from plasma nonuniformity. As the charge builds up on the wafer, it may not be uniform because electron shading can also take place because of the surface features blocking the charge to some areas of the wafer, resulting in a charge differential across the wafer surface and across the insulating gate dielectric. When the charge build-up on the sensitive gate dielectric is sufficiently high, it

may cause electrical damage, such as charge trap generation or dielectric breakdown [185].

Tests have been developed to quantify charging damage, typically based on breakdown of the sensitive gate oxide. Antenna structures [186] have shown that plasma nonuniformity contributed to charging of the gate surface and thus degradation of the gate oxide. Models for damage in stripping systems are based on plasma ion density and plasma uniformity; in addition, the effects of substrate doping concentration on plasma-induced oxide damage have been investigated by You et al. [187]. Furthermore, wafer substrate temperature is a key parameter with respect to plasma charging during the resist stripping process [188].

Plasma-related damage is observed at various steps of the device manufacturing process. Plasma-induced substrate damage occurs during high-dose implant resist strip processes. Chan et al. [189] observed device damage at the HDIS step and controlled this damage by optimization of the source (ICP) and bias (RF) power, while Shin et al. [190] observed damage to the thin gate oxide at the Al etching and stripping process steps and showed that the plasma stripping step accounted for more damage than did the etching step [191].

7.7 CONCLUSIONS

Plasma stripping, cleaning, and surface conditioning processes are widely used in IC manufacturing. Both remote downstream or remote plasma and RF-assisted cleaning processes are common and both yield good results with respect to surface cleanliness.

The development of new plasma sources will undoubtedly occur, especially for the more advanced technologies, which requires surface state and charging control. For example, the use of neutral beams or neutral active reactant flux generated from neutralization of ions from plasma sources offers the potential to minimize plasma damage while creating directionality which, with appropriate energy control, may allow improved selectivity without charge impingement on surfaces [192–194]. New plasma chemistries are continuously being evaluated, along with optimization of the process parameters to yield lower cost processes with better residue cleaning capabilities.

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